

Anesthesia unconsciousness recruits dynamic parabrachial activity within a distributed brain-wide ensemble

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SUMMARY

Neural circuits underlying unconsciousness remain poorly defined. We test the hypothesis that unconsciousness arises from specific, distributed circuits using general anesthesia in mice as a

39 reproducible model. We identify a cortical-to-subcortical shift in neural activity during isoflurane
 40 anesthesia that is organized into nine discrete functional communities mapped at single-cell
 41 resolution. The lateral parabrachial nucleus (LPB) emerges as a central hub, exhibiting high
 42 interconnectivity, spontaneous firing under anesthesia, and preferential recruitment during
 43 reactivation of the brain-wide ensemble confirmed by single unit recordings. Chemogenetic
 44 reactivation of the captured brain-wide ensemble induces sedation, slow wave oscillations,
 45 hypothermia, and analgesia, which are components of anesthesia-induced unconsciousness.
 46 Reactivation of the LPB ensemble alone recapitulates a subset of these effects. Together, we define
 47 a global neural substrate for unconsciousness and recapitulate its dissociable autonomic,
 48 neurophysiologic, and behavioral effects using brain-wide ensemble manipulations. These results
 49 establish a neural circuit framework for anesthesia-induced unconsciousness in the mammalian
 50 brain.

51

52 **KEYWORDS (UP TO 10 KEYWORDS)**

53 anesthesia, analgesia, unconsciousness, TRAP, chemogenetic, light sheet microscopy, Neuropixels,
 54 Fos, lateral parabrachial nucleus, isoflurane

55

56 **HIGHLIGHTS** (up to 85 characters each)

- 57 - A distributed brain-wide ensemble comprises anesthesia unconsciousness
- 58 - Lateral parabrachial nucleus is a key, preferentially activated hub
- 59 - Brain-wide or parabrachial ensemble re-activation induces altered consciousness
- 60 -

61 **INTRODUCTION**

62 A prevailing hypothesis that general anesthesia produces unconsciousness by targeting a specific,
 63 controllable neural circuit substrate is yet to be confirmed. Many spatially segregated neural circuits
 64 are shown to regulate consciousness or recapitulate distinct components of an unconscious state^{1,2}.
 65 General anesthesia likely encompasses neural circuit components of analgesia, sedation,

immobility, and amnesia that together produce loss of consciousness³. Dissecting individual cell types and circuits contributing to these components has yielded valuable insights into the regulation of sleep^{2,4-8}, arousal⁹⁻¹⁶, and anesthesia¹⁷⁻²⁵, though with limited translatability for directly controlling consciousness.

The wet blanket explanation of anesthesia hypothesizes that anesthetic drugs work non-specifically to dampen excitatory signaling and enhance inhibitory signaling to generate unconsciousness²⁶, or that anesthesia modulates mitochondrial metabolism in broad cell types²⁷. Alternative hypotheses posit that specific neural circuits mediate anesthesia-induced unconsciousness²⁸, such as specific cortical layer 5 decoupling^{20,29}, or that anesthesia hijacks endogenous sleep and arousal-regulating pathways to produce unconsciousness^{2,4}. Chemogenetic or optogenetic manipulation of cortical, thalamic, and hypothalamic nuclei, including the median preoptic nucleus (MEPO)³⁰, as well as pharmacologic GABA_A-R agonist infusions into the mesopontine tegmental anesthesia area (MPTA)³¹, have established multiple neural circuits important for loss-of-consciousness. A single, isolated neural correlate of unconsciousness has not been identified³². Here, we directly test the hypothesis that anesthesia-induced unconsciousness is subserved by globally distributed neurons in targetable circuits^{33,34}. If this is true, a distributed, specific neural substrate of the anesthetized unconscious state exists that can be manipulated to modify conscious state transitions.

We apply a preclinical approach using intact-brain light-sheet fluorescent imaging³⁵⁻³⁷, activity-dependent targeted recombination in active populations (TRAP)^{38,39}, chemogenetic activation⁴⁰, high density single unit electrophysiology⁴¹, wireless mechano-acoustic recording of internal physiology⁴² and machine learning behavioral classification^{43,44} to manipulate and analyze cell-specific activity within the brain-wide isoflurane anesthesia unconsciousness network in mice.

Our approach better informs the neural framework of unconsciousness, as defined by general anesthesia, and investigates avenues for developing control of consciousness by targeting neural circuits.

While the parabrachial nucleus has an established role in arousal^{14,45–48}, little is known about its contribution to generating unconsciousness. Our single-unit recordings across the brain indicate preferential recruitment of hindbrain activity in the brain-wide anesthesia ensemble that is driven by a lateral parabrachial (LPB) hub, also identified as a key community in the Fos activity map. We manipulate the LPB neural ensemble, revealing similar patterns of behavior as seen with brain-wide reactivation, including analgesia, freezing, hypothermia, and potentiation of the loss of righting reflex. Together, these studies establish a critical role for anesthesia-activated LPB ensembles in the unconscious state transition, as well as define the associated cellular resolution brain-wide network.

RESULTS

Single-cell resolution brain-wide activity map of isoflurane unconsciousness defines a cortical-to-subcortical shift in activity

To identify the neural circuitry of anesthesia-induced unconsciousness, we constructed an intact brain-wide map of activity following isoflurane (ISO) anesthesia using immediate early gene immunolabeling and light-sheet microscopy. Mice were anesthetized and maintained under 1.2–1.3% ISO for 180-min prior to fixation and intact brain processing³⁷ (**Fig 1A**). Control mice were singly housed in the home cage. We analyzed whole-brain imaging datasets using two complementary pipelines, ClearMap⁴⁹ and UNRAVEL³⁶, which provide orthogonal views of Fos

112 activity (**Fig 1B-C** and **Fig S1-S2**). ClearMap uses a cell-detection-based strategy to quantify Fos+
 113 puncta after registration, producing both regional cell-density measures and voxelized statistical
 114 maps. In contrast, UNRAVEL prioritizes voxel-wise mapping of statistically robust differences in
 115 Fos immunolabeling intensity after background subtraction and z-scoring, then characterizes the
 116 anatomical composition and Fos+ cell densities of atlas-agnostic clusters of significant voxels (see
 117 **Methods** for a more detailed discussion). We followed these complementary analyses with network
 118 and community analysis of regional Fos maps (**Fig 2**)⁵⁰.

119 Using both approaches, we observe differential Fos patterns in the ISO state relative to
 120 control mice, with a shift from cortical to subcortical activation in the ISO-induced unconscious
 121 state (**Fig 1C**, **Fig S1B**, **Table S1**). When analyzed at a gross hierarchical level, the total mean Fos
 122 density is consistent across analyzed samples, significantly decreases with ISO in the isocortex and
 123 midbrain, and increases in the hypothalamic areas and cerebellum (**Fig S1G**; see **Fig S1C-D** for
 124 representative Fos images). At the granular level, ISO increases activity within subregions of the
 125 striatum (caudoputamen and central amygdala), pallidum (bed nucleus of the stria terminalis),
 126 thalamus (paraventricular nucleus), hypothalamus (paraventricular and ventromedial nuclei), and
 127 hindbrain (lateral parabrachial nucleus, locus coeruleus, and reticular nucleus) (**Fig S1H**).
 128 UNRAVEL confirmed these increases in activity (**Fig 1E-G**) and extended the findings to identify
 129 significant ISO-activated subcortical clusters that encompass multiple discrete brain subregions (see
 130 **Video S1** and **Fig S2**). For example, cluster 3 is predominantly in the LPB and neighboring pontine
 131 regions. Hotspots and coldspots from ISO-treatment are summarized in **Table S1**. Together, this
 132 shows that ISO activation occurs across several subcortical brain regions, with an overall cortical to
 133 subcortical shift in activity. These data establish a brain-wide neural framework for interpreting
 134 ISO-induced unconsciousness (**Fig 1**) and enable further network analysis (**Fig 2**).

135

136 ***Network analysis of Fos maps reveals high interconnectivity of distributed communities***

137 Since a given brain state is composed of dynamic interconnectivity between brain regions,
 138 applying network analysis approaches to study the structure of connectivity (i.e., network topology)
 139 can generate insights about the functional patterns that differentiate behavioral states^{51–53}. We
 140 leveraged our brain-wide Fos mapping datasets to investigate how functional connections are
 141 changed by ISO-induced unconsciousness (**Fig 2**) using SMARTTR⁵⁰, a software package
 142 supporting independently mapped IEG datasets for network analysis and visualization. We cross-
 143 correlated area-normalized Fos across all mapped subregions (**Fig 2A-B**) and found a higher
 144 number of strongly positive correlations in the isoflurane condition (**Fig S3**). Since Fos is an
 145 activity-dependent marker, positive correlations between brain regions represent either co-
 146 activation or co-suppression, while negative correlations represent opposing activation.

147 To better investigate which individual connections are most functionally altered by ISO, we
 148 subtracted the strength of each regional connection (i.e. correlation coefficient) in the control
 149 condition from its corresponding value in the ISO condition and compared each difference to a
 150 permuted null distribution (**Table S2**). Consistent with our cross-correlation analysis, there was a
 151 greater proportion of positively shifted connections in ISO (**Fig 2C, Fig S3**). We next constructed a
 152 network of the most significant functionally altered connections ($p < 0.001$), visualizing each
 153 correlation difference as an edge (lines), and the regions they connect as nodes (circles) (**Fig 2D**).
 154 This network revealed a naturally high interconnectivity amongst regions involved, visualized as a
 155 large, connected component (i.e., sub-network), rather than as collections of small, isolated
 156 networks or individual connections, as would be expected from chance (**Table S2**).

157 We also constructed individual networks (edge proportion 3.15%), to examine more general
 158 topological properties (**Fig 2C, Fig S3**). In the context of the brain, the property of small-
 159 worldness⁵⁴ can be considered a balance between the capacity for functional specialization and
 160 integration. We confirmed that both the control and ISO networks exhibited small-worldness,
 161 defined as $\sigma > 1$, although σ was lower in ISO (3.61) compared to control (5.67) (**Fig S3D**). We
 162 examined global network properties across a range of alpha thresholds and proportional thresholds,
 163 finding the density of connections in the ISO network is consistently higher across a range of alpha
 164 thresholds, reflected in the higher trajectories of mean degree, clustering, and efficiency (**Fig S3B**).
 165 This effect is mitigated when controlling for edge density (**Fig S3C**). As a result, we found that
 166 global topological properties are highly influenced by greater functional interconnectivity in the
 167 isoflurane unconsciousness network.

168 Many complex real networks can be broken into subparts called communities, defined by
 169 nodes with high interconnectivity or probability of interconnectivity to other members within their
 170 community. We used an agglomerative community detection algorithm⁵⁵ and discovered nine
 171 distinct communities within our functional difference network (**Fig 2D-E**). Metrics of nodal
 172 influence on a network, termed centrality, include degree (the number of direct connections to a
 173 node) and betweenness (how often a particular node lies on the shortest path between two other
 174 nodes). Examination of degree and betweenness (**Table S2**) revealed higher centrality of regions
 175 such as the MEPO, microcellular tegmental nucleus, lateral mammillary nucleus (LM), locus
 176 coeruleus, and LPB. Qualitatively, these metrics align well with the central, influential positioning
 177 of these regions within distinct, spatially distributed communities (See **Video S2**). Together,
 178 community detection identified nine distinct communities. The LM, MEPO and LPB hub

179 communities were the most densely interconnected and covaried significantly with the ISO
180 unconscious state (**Fig 2D**).

181

182 ***Integrating behavioral analysis with internal dynamics during conscious state transitions to***
183 ***evaluate neural ensemble contributions***

184 To investigate the contribution of this identified brain-wide network to ISO-induced conscious state
185 transitions, we established an awake, behaving platform for concurrent assessment of dynamic
186 internal physiology with automated behavioral classification during state transitions (**Fig 3**). We
187 subdermally implanted mice with wireless, battery-free mechano-acoustic devices that detect heart
188 rate, respiratory rate, physical activity, and body temperature, and that we previously validated for
189 use with ISO⁴² (**Fig 3A**). Mice were then placed individually into a small modified open field
190 chamber (15x15cm) that also permits ISO administration (**Fig 3A-B**). Behavioral state transitions
191 were quantified with machine-learning-based supervised behavioral classification using pose
192 estimation to capture rearing, grooming and immobility bouts in an automated, objective manner
193 (**Fig 3C**)^{43,44}. We established that the wireless mechano-acoustic devices capture a reproducible
194 decrease in heart rate, respiratory rate, temperature and physical activity while mice underwent
195 induction of ISO anesthesia (**Fig 3D, Video S3**) and their pose trajectories reflected reduced
196 locomotor activity (**Fig 3E**). Using this behavioral platform, we went on to test the contribution of
197 brain-wide ensembles to anesthesia-induced unconsciousness.

198

199 ***Brain-wide activity-dependent capture of anesthesia unconsciousness***

200 We hypothesized that if unconsciousness is regulated by discrete brain-wide-activated neurons, it
 201 should be recapitulated by activating the brain-wide neural ensemble. To capture brain-wide
 202 anesthesia-activated unconsciousness circuitry, Fos-TRAP2 (Fos2A-iCreER)³⁸ mice received
 203 retroorbital injections of AAV-PHP.eB virus expressing Cre-dependent designer receptors
 204 engineered to be activated by designer drugs (DREADDs)⁵⁶. After 3-4 weeks, mice underwent 180-
 205 min of ISO exposure (1.2-1.3%) and halfway through the exposure, at 90 min, received an
 206 intraperitoneal (i.p.) injection of 4-hydroxytamoxifen (4-OHT) to induce activity-dependent
 207 DREADD expression in brain-wide ISO-activated neurons (TRAP)^{38,39}. On separate subsequent
 208 days, mice were implanted with either electrodes for electrocorticography (ECoG) or a wireless
 209 mechano-acoustic device⁴² to characterize neurophysiological and peripheral physiological effects,
 210 respectively, as well as behavioral changes after chemogenetic manipulation (**Fig 4A**).

211 After chemogenetic induction with the DREADD ligand clozapine-n-oxide (CNO, 5mg/kg
 212 i.p.), Gq-DREADD-expressing mice display increases in slow wave delta (1-4 Hz) oscillations as
 213 compared to their saline trials (**Fig 4B-C**, see **Fig S4** for spectrograms). We highlight a “drug
 214 onset” dashed line where the average onset of behavioral sedation occurs across CNO trials,
 215 approximately 5-10 minutes following injection. CNO induction promotes a significant increase in
 216 slow wave (1-4 Hz) signal to noise ratio similar to ISO induction (**Fig 4B-C**). Group-averaged
 217 spectrograms analyzed pre- and post-CNO exposure show this similarity between CNO and ISO
 218 induction, as well as a distinction from saline (**Fig S4B**).

219 Anesthesia-induced unconsciousness in rodents is commonly defined by loss of righting
 220 reflex (LORR)⁵⁷, a postural reflex that allows rodents to regain the upright position. We find that
 221 CNO potentiates LORR with subanesthetic ISO administration, though LORR does not occur at
 222 baseline (**Fig 4E** and **Fig S4C**). A significant shift toward LORR occurs after chemogenetic

stimulation of ISO-activated brain-wide circuitry in both CNO doses tested (5 and 10mg/kg) (**Fig S4A**). LORR potentiation does not occur in non-TRAP control conditions (**Fig S4C**) or in Gi-DREADD-inhibition of the brain-wide ensemble (**Fig S4F**). CNO (1 and 5 mg/kg) also produces dose-dependent reductions in movement velocity in activating Gq-DREADD-expressing mice (**Fig S4D**) but not inhibitory Gi-DREADD-expressing mice (**Fig S4G**). In behavioral tests, we continue to use 5mg/kg CNO, as this dose is validated to saturate DREADD receptors with minimal off-target effects^{58,59}, does not modify behavior when administered to non-TRAP control and Gi-DREADD-expressing mice, and does not fully immobilize Gq-DREADD-expressing mice, allowing us to analyze behavioral repertoires of this altered state of consciousness.

We went on to examine behavioral effects after chemogenetic manipulation of the brain-wide ensemble in another cohort of mice. We assayed if coordinated movement, as measured in the rotarod test, is affected by chemogenetic stimulation of brain-wide anesthesia-activated circuitry. Mice underwent 2 days of training and were then placed on a slowly rotating rod (4 revolutions per minute, rpm) that poses minimal challenge to motor coordination under control conditions. After clozapine-n-oxide (CNO) injection (5 mg/kg), mice expressing activating Gq-DREADD fell off the rotarod when compared to their respective saline trials, indicating a loss of coordinated movement (**Fig 4D**), which is not observed in mice expressing brain-wide inhibitory Gi-DREADD (**Fig S4M**).

We observed that chemogenetic stimulation of ISO-activated brain-wide circuitry produces distinct behaviors, with differing temporal onset and distributions, when mice are tested and recorded in the modified open field chamber in the absence of anesthetic drug. An example raster plot of 6 cumulative saline and CNO trials collected from one Gq-DREADD-expressing mouse shows the representative distribution of 5 behaviors before and after drug onset (**Fig 4F**, also see **Video S4**). Using machine-guided supervised behavioral classification to quantify behavior relative

246 to control saline injections, we find that CNO induction results in decreased rearing and grooming
247 behavior, concomitant with an increase in stereotyped circling behavior, Straub tail, and freezing
248 (**Fig 4G**). These effects were reproducible with each CNO trial across 4 separate cohorts of mice
249 tested (see **Table S3** for detailed statistics).

250 Since altered consciousness may be associated with widespread physiological changes⁶⁰, we
251 analyzed peripheral physiological data collected from wireless mechano-acoustic devices that we
252 previously validated for use under ISO. We aligned behavioral and physiological data to the
253 average latency to the first bout of freezing observed across CNO induction trials (5 min after CNO
254 injection, “drug onset”). Consistent with an altered state of consciousness, we observe significant
255 decreases in body temperature and heart rate recorded after CNO induction (**Fig 4H-I**). CNO also
256 increased mean respiratory rate in Gq-DREADD mice (**Fig S4E**).

257 In contrast, we did not see significant behavioral changes, besides decreased rearing, in
258 brain-wide Gi-DREADD-expressing mice (**Fig S4H-I**). Mice that express Gi-DREADD never
259 show the atypical behaviors, such as Straub tail, seen in the Gq-DREADD mice (**Fig S4H-I**). They
260 also retain stable peripheral physiology after CNO induction (**Fig S4J-L**).

261 Since anesthesia-induced unconsciousness can encompass analgesia, we sought to identify
262 the anti-nociceptive function of chemogenetic stimulation. Mice were tested for thermal anti-
263 nociceptive responses in the warm water tail withdrawal (tail flick) test interleaved with hotplate
264 testing after saline and at 30-min time points after CNO injection (**Fig 4J**). In the tail flick test,
265 chemogenetic activation promoted significantly delayed tail withdrawal (**Fig 4L**). In hotplate
266 testing, chemogenetic activation reduced jumping off the hot plate, decreased paw withdrawals, and
267 increased latency to first response after CNO (**Fig 4M**). There was no effect of chemogenetic

manipulation on thermal anti-nociceptive responding in non-TRAP control or TRAP Gi-DREADD mice (**Fig 4L-M**).

Following all behavioral tests, intact brains were cleared and immunolabeled for mCherry+ DREADD expression using iDisco+ and light-sheet microscopy to quantify brain-wide DREADD distribution (**Fig 4K** and **Fig 5**). Globally, we observe consistent brain-wide TRAP2 cell population density across gross cortical and subcortical divisions, besides significant differences in the hindbrain and cerebellum across samples analyzed (**Fig 5B**). To better understand brain region-specific differences, we used a voxel-based analysis where 3D and 2D TRAP2 density heatmaps exhibit clear patterns of region-specific TRAP2 enrichment (**Fig 5C-D**, **Table S3**). These expression patterns are not seen in non-TRAP controls that received oil vehicle instead of 4-OHT during TRAP (**Fig 5E-F**).

By capturing the brain-wide anesthesia-activated neural ensemble, we describe the chemogenetic induction of an altered state of consciousness that mirrors patterns of behavioral, neural and physiological activity observed in natural sleep⁶¹, torpor^{62,63}, and anesthesia⁶⁴. This altered state is characterized by a significant increase in 1-4Hz slow wave oscillations, atypical freezing behavior, hypothermia, loss of coordinated movement, and thermal anti-nociception. Chemogenetic induction also potentiates LORR but does not generate LORR in the absence of anesthetic.

Preferential single unit recruitment of isoflurane-responsive ensembles in the parabrachial nucleus of the hindbrain

Our previous experiments evaluated brain-wide cellular activation derived from static snapshots of Fos and DREADD expression maps. To examine the temporal dynamics of neurons engaged by

291 ISO anesthesia, we used Neuropixels probes to record single-unit activity across multiple brain
292 regions as mice underwent ISO exposure and chemogenetic reactivation of the captured brain-wide
293 ensemble (**Fig 6**). Briefly, mice underwent surgery and were habituated to a custom head-fixed rig
294 to facilitate acute Neuropixels recordings⁴¹ (**Fig 6A**, see **Methods** for details). In each session, we
295 recorded activity starting with a 5-min awake baseline before ISO induction, after which mice were
296 maintained unconscious under 1.3% ISO for 20-min (**Fig 6A**, “ISO”) before drug washout (25-
297 min). After a second 5-min awake baseline, mice received CNO (5 mg/kg, i.p.) to re-active brain-
298 wide neural ensembles (**Fig 6A**, “CNO”, 30 min). Dye-labeled probe trajectories were traced in
299 light sheet images obtained from the intact, cleared brains of recorded mice (**Fig 6B**). A total of
300 2076 units from 8 mice were isolated and included for analysis (**Fig 6C**).

301 To determine whether CNO re-engaged the same units activated by ISO, we first classified
302 each recorded unit by its responsiveness to ISO and CNO reactivation separately and evaluated
303 their overlap within each region. We analyzed brain-wide responses across recorded regions and
304 CNO-responsive proportions (**Fig S5**, **Table S4**). Across brain regions, CNO responses did not
305 always maintain their original ISO-activated or suppressed response direction. Among units
306 responsive to both ISO and CNO, a majority exhibited directional agreement. However, their
307 proportion did not exceed chance in any region and same-direction pairs were overwhelmingly of
308 joint suppression (**Table S4**). Thus, reactivation of the brain-wide ensemble re-engages units
309 captured during ISO anesthesia and partially restores their anesthesia-state firing patterns.

310 At the population level, both major divisions, isocortex (n = 346) and hindbrain (n = 361),
311 contained significant proportions of ISO-responsive and CNO-responsive units, exceeding shuffle
312 null rates for each region ($q < 0.001$; **Fig 6D-E**, **Table S4** for detailed statistics). Responsive units
313 were strongly biased toward suppression under ISO and CNO alike, consistent with broad inhibition

that characterizes the anesthetized and chemogenetic re-activated states. Despite a shared suppressive signature, cortex and hindbrain diverged in whether CNO preferentially re-recruited the ISO-responsive ensemble (**Fig 6F, 6H**). In isocortex, CNO-responsive units were no more likely to be ISO-responsive than CNO-non-responsive units ($OR = 0.84$, $q = 0.82$), indicating that reactivation engaged cortical units independently of their past ISO responsiveness. In hindbrain, by contrast, ISO-responsiveness was significantly enriched among CNO-responsive units ($OR = 2.15$, $q = 0.007$), indicating preferential re-recruitment of the same units engaged during ISO anesthesia (**Fig 6F**). Together, this positions the hindbrain as preferentially re-activated relative to isocortex.

To determine whether preferential recruitment of the hindbrain was distributed across the region or localized to a specific node, we separated the LPB ($n = 95$), identified as a key hub in our whole-brain network analysis, from the surrounding non-PB hindbrain (nonPB-HB, $n = 266$). LPB showed significant preferential re-recruitment (**Fig 6G**), with CNO-responsive units enriched for ISO-responsiveness relative to CNO-non-responsive units ($OR = 2.65$, $q = 0.036$). The surrounding non-PB hindbrain did not reach significance ($OR = 1.60$, $q = 0.17$). Directly comparing the two subdivisions (CNO-responsiveness by region interaction) did not reach significance (Firth-penalized likelihood-ratio test, interaction $OR = 1.60$, $p = 0.18$), showing an effect in the predicted direction of LPB's relatively greater re-recruitment (**Fig 6I** see also **Fig S5G, Table S4**). These results are consistent with the preferential single unit recruitment of a parabrachial ensemble that includes LPB as a key hub.

333

Lateral parabrachial ensembles recapitulate altered consciousness when chemogenetically manipulated

336 Across our analyses, we discovered prominent activation of the LPB. LPB is significantly activated
 337 by ISO in Fos whole brain activity maps (**Fig 1**), emerges as a highly interconnected community
 338 (**Fig 2**) in network analysis of the Fos maps, exhibits neuronal firing under anesthesia and is
 339 preferentially recruited with chemogenetic stimulation after brain-wide activity-dependent capture
 340 (**Fig 6**). We hypothesized that if unconsciousness is regulated by discrete neural circuitry in the
 341 global network, unconsciousness should be recapitulated by activating highly significant,
 342 interconnected neural ensembles in the network, like the LPB. To capture ISO-activated LPB neural
 343 ensembles, we intracranially injected the same AAV-PHP.eB virus expressing Cre-dependent
 344 DREADD locally in the LPB of FosTRAP2 mice and subjected them to the same TRAP procedure
 345 under ISO, followed by behavioral testing with implanted wireless MA devices (**Fig 7A**). After
 346 behavioral tests, brains were processed with immunohistochemical labeling of mCherry-DREADDs
 347 observed predominantly at the LPB injection site, with some neurons also captured in pontine
 348 nuclei surrounding the injection site (**Fig 7B** and **Fig S6L-M**). Mice underwent alternating trials of
 349 saline and CNO with concurrent videorecording in the modified open field box. We quantified
 350 behavior and peripheral physiology using the same automated classification pipeline as with brain-
 351 wide ensemble chemogenetic manipulations to evaluate the contribution of the LPB ensemble to
 352 altered consciousness. A representative plot showing the behavioral classification from the average
 353 of one animal's saline and CNO trials is shown in **Fig 7C** (also see **Video S5**).

354 Similar to brain-wide ensemble manipulations, we observed a significant reduction in
 355 rearing and grooming behaviors after CNO induction, as well as prominently reduced locomotor
 356 movement (**Fig 7D**, **Fig S6**). We applied the same heuristic approach to quantify reduced
 357 movement bouts, finding that LPB ensemble stimulation induced significant freezing (**Video S5**).
 358 LPB ensemble re-activation did not induce circling bouts that were observed after brain-wide

ensemble stimulation (**Fig 4G**), rather circling behaviors significantly decreased (**Fig 7D**). In peripheral physiological recordings, similar patterns of decreased heart rate and decreased temperature were induced (**Fig 7E-F**) with no significant change in respiratory rate after chemogenetic stimulation (**Fig S6H**). These behavioral and physiological effects were not observed in mice expressing inhibitory Gi-DREADDs in the LPB anesthesia-activated ensemble (**Fig S6E-G, S6I-K**). Overall, these data support the causal role of an LPB hub in generating unconsciousness.

CNO again potentiated loss of consciousness as defined by LORR, with LORR occurring at significantly lower ISO concentrations compared to the animal's saline trials in Gq-DREADD mice (**Fig 7G**). However, LORR was not generated by stimulating the LPB neural ensemble in the absence of ISO. No significant potentiation of LORR by ISO was observed in Gi-DREADD mice (**Fig S6C**). After CNO injection (5 mg/kg, i.p.), neither Gq-DREADD nor Gi-DREADD mice showed a change in latency to fall off the rotarod when compared to their respective saline trials, indicating no changes in coordinated movement (**Fig S6A, D**). Given the established contribution of LPB circuitry to pain regulation, we also assessed the effects of chemogenetic induction on analgesia using the tail flick test. We observed significantly increased latency to tail flick in Gq-DREADD-expressing, but not Gi-DREADD mice, indicating that LPB ensemble stimulation, but not inhibition, promotes thermal anti-nociception (**Fig 7H**). Taken together, these data show that stimulation of the ISO-activated LPB neural ensemble is sufficient to recapitulate dissociable components of altered consciousness observed after brain-wide ensemble manipulation, establishing a causal contribution of LPB neural activation within the broader anesthesia network.

The LPB is a heterogeneous brain region composed predominantly of glutamatergic neurons with diverse transcriptional profiles, as well as smaller GABAergic subpopulations⁶⁵. To determine

if the ISO-activated LPB ensemble was dominated by a specific marker-defined neuronal subpopulation, we integrated ISO Fos maps (**Fig 1D-G**) with Allen Brain Cell Atlas MERFISH spatial transcriptomic data and quantified imputed neurotransmitter, LPB clade, and neuropeptide marker expression within the LPB portion of cluster 3 (**Fig 7I, Fig S7, and Table S5**). Fos-cluster neurons mapped onto diverse LPB populations, including *Atoh1*-related and *Lmx1*-related glutamatergic clades and sparse GABAergic populations (**Fig 7I and Fig S7**). Neuropeptides and cell-type-specific markers showed diverse expression patterns across LPB cluster neurons (**Table S5**). For example, *Pdyn* and *Cck* were enriched in the *Meis2*⁺ *Atoh1*-related clade, which comprised 58.4% of LPB cluster neurons, whereas *Calca*, *Nts*, and *Tac1* were enriched in the *Lmx1b*⁺ *Lmx1*-related clade, which comprised 27.0% of LPB cluster neurons (**Fig S7**). Together, these data indicate that no single marker-defined LPB cell type exclusively comprises the ISO-activated neural ensemble. Nearly all mapped glutamatergic LPB projection targets⁶⁵ also showed increased Fos after ISO exposure, consistent with broader downstream network recruitment (**Fig S7**). In conclusion, the LPB ensemble is heterogenous and not defined by a single cell-type or subregion that underlies unconsciousness.

397

DISCUSSION

In this series of experiments, we characterize the brain-wide ISO anesthesia unconsciousness network at cellular resolution. We test the hypothesis that neural circuit substrates of unconsciousness exist that can be selectively manipulated using activity-dependent chemogenetic manipulation to modify consciousness. We conclude that altered consciousness is recapitulated by manipulating either the brain-wide neural ensemble or an LPB hub, establishing a distributed neural circuit framework for anesthesia unconsciousness.

Recent circuit dissection studies in rodents have focused on stimulating or inhibiting isolated brain nuclei to uncover the contributions of individual brain regions to unconsciousness. Many studies have focused on hypothalamic nuclei that we also identify as significant, such as the preoptic nucleus. While these studies provide insights into region-specific contributions, they have also produced paradoxical findings. For example, targeted stimulation of MEPO contributes to arousal from anesthesia despite it also having high activity during sleep and anesthesia maintenance³⁰. Selective stimulation of either GABA or glutamate subpopulations in the preoptic nuclei is insufficient for generating unconsciousness⁶⁶. One potential explanation for these differences is that cell populations necessary for unconsciousness are functionally rather than genetically distinct, as we found in LPB, and activity-dependent manipulations are necessary to disentangle their contributions.

In addition to LPB, our whole brain connectome also identifies the MEPO as a key community node contributing to the unconscious state. The preoptic nucleus of the hypothalamus and the LPB may both contribute to hypothermia and coordinate autonomic responses^{62,67–70}. The central amygdala is another community node that was found previously to modulate pain responding under anesthesia¹⁸. Since significant data exists to support causal contributions of hypothalamic^{15,16,30,66,71,72} and amygdala nuclei^{18,73,74} to anesthesia-induced unconsciousness, we focused our region-targeted experiments on LPB.

Previous studies have shown that stimulation of the parabrachial nucleus accelerates arousal from an unconscious state^{45–47}. However, these prior studies were performed without activity-dependent stimulation and may recruit populations in the region that promote arousal. Spatially aligning our Fos maps with the Allen Brain Cell Atlas shows that the ISO-activated LPB cluster encompasses diverse LPB cell-types, including glutamatergic cells expressing calcitonin gene-

related peptide (CGRP), as well as GABAergic populations⁶⁵. These LPB cell-types have distinct, bidirectional effects on freezing behavior and pain responding, where glutamatergic CGRP-positive cells increase freezing⁷⁵ and induce nociplastic pain⁷⁶, while GABA cells alleviate pain when stimulated⁷⁷. Our behavioral findings indicate that recruitment of both LPB cell populations likely occurs after activity-dependent capture, with GABA stimulation promoting anti-nociception and glutamatergic stimulation promoting freezing behavior, as previously reported in the literature. This divergence reflects the complexity of behaviorally defining heterogeneous neural interconnectivity and underlines the importance of harnessing integrated, activity-dependent and systems-level investigations for delineating neural mechanisms of unconsciousness.

Taken in context with the known effects of general anesthesia on cortical layer 5 pyramidal cells^{7,20,29,78}, the distributed, LPB-centered network described here may be upstream of cortical cellular-level effects and reflect different levels of a shared process. Whether reactivating the anesthesia unconsciousness ensemble, or the LPB hub specifically, alters cortical coupling is an open and testable question to link these circuit and cellular mechanisms.

Analysis of single unit firing patterns indicates that a recruitment of ISO-activated neurons occurs with chemogenetic stimulation, particularly in the hindbrain. Many neurons exhibit similar suppressed firing with CNO induction, but ISO-activated neural firing patterns are different. This suggests either ISO recruits additional neuronal or non-neuronal mechanisms to generate unconsciousness, or the ensemble captured during ISO maintenance diverges from neurons necessary for induction of unconsciousness⁷⁹.

Surprisingly, we find that neural ensemble activation alone does not generate LORR, reflecting a limitation of activity-dependent chemogenetic stimulation, or suggesting non-neuronal or non-specific, wet-blanket mechanisms are necessary for this loss-of-consciousness

behavior^{27,57,80–82}. Activating brain-wide neural ensembles induces an altered consciousness that is reflected by behavioral sedation and increased slow wave oscillations but not captured by LORR as a behavioral proxy. The slow wave state produced is comparable to a non-REM sleep pattern², further supporting a convergence of anesthesia and sleep neural circuitry. Similarly, LPB ensemble activation potentiates LORR, though it does not generate it without ISO administration. The LORR behavioral definition is not applicable to altered states of consciousness like sleep and psychedelic states^{83,84} suggesting chemogenetic re-activation of anesthesia neural circuitry more closely resembles other altered states of consciousness.

Thermal anti-nociception accompanying the chemogenetic state transition may be confounded by and reflect the altered state of consciousness. We previously showed that anti-nociception can be recruited at lower CNO doses than those necessary for immobility after brain-wide activity-dependent manipulations using a different transgenic approach⁸⁵, mimicking drugs that have analgesic effects at sub-anesthetic doses. Thus, brain-wide activity-dependent capture also provides a platform for dissociating neural circuit components of analgesia and unconsciousness circuitry, if such dissociations exist. Developing a deeper understanding of unconsciousness neural circuitry could inform therapeutic approaches, such as focused ultrasound⁸⁶ or brain stimulation approaches⁸⁷, to modify pain, sleep, coma and disorders of consciousness.

468

469 *Limitations of the study*

While the TRAP2 approach is highly specific (>95% of activity-labeled cells are Fos+), it is only moderately efficient (~65% of Fos+ cells are activity-labeled)⁵⁸. Similarly, following retro-orbital injection at viral titers like our own ($2-3 \times 10^{11}$), PHP.eB neuron infection efficiency is high in the cortex (~70% of neurons are infected), but moderate in other brain areas like the striatum

(~50% of neurons)⁴¹. Though these transgenic and viral efficiencies are sufficient for inducing significant behavioral changes, they introduce technical limitations for analyzing endogenous brain-wide mechanisms, so we also use endogenous Fos mapping. Our behavioral readouts are also limited by the temporal expression of Fos for activity-dependent capture, and the efficiency of CNO or DCZ to induce neural depolarization via DREADD receptor activation. We previously used a TRAP2 chemogenetic approach that uses transgenic DREADD expression independent of viral efficiency and report less robust behavioral results of brain-wide activation⁸⁵. To account for temporal limitations of Fos as a readout of anesthesia activation, we also use single unit neurophysiology.

483

484 **RESOURCE AVAILABILITY**

485 ***Lead contact***

486 Requests for further information and resources should be directed to and will be fulfilled by
487 the lead contact, Mitra Heshmati (mhesh@uw.edu).

488

489 ***Materials availability***

490 This study did not generate new unique reagents.

491

492 ***Data and code availability***

493

- 494 All original code for behavioral analysis is available at <https://github.com/sgoldenlab/simba>
- 495 All original code for UNRAVEL analysis is available at
496 <https://zenodo.org/records/13655988>.
- 497 All code for SMARTTR analysis is available at <https://mjin1812.github.io/SMARTTR>
- 498 Datasets are included in compiled supplemental tables.
- 499 Any additional information required to reanalyze the data reported in this paper is available
500 from the lead contact upon request.

501

502

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AUTHOR CONTRIBUTIONS

MH and SAG conceptualized and directed all aspects of the study. MH, SAG, KNS, MHCN, DRR and MJ wrote the first draft of the manuscript. SAG, MH and MRB designed whole brain studies. KNS, MHCN, HL, DRR, JN, ADM, ACB, IC, MH and SAG collected, processed and/or imaged brain tissue. DRR, ADM, ES, IC, KI, GS, BDH, SAG and MH performed whole brain analysis. MJ and CAD performed network analysis. JN and ES prepared viruses and performed retroorbital virus injections. MHCN, HL, CN, ACB and MH performed activity-dependent chemogenetic behavioral experiments. MHCN, HL, NR and AL collected and analyzed mechano-acoustic recordings. SRON, HL, MHCN, CCS, NR, EA, JZ, and AL analyzed behavioral video recordings. KNS performed and analyzed high density silicon probe experiments. AB, GG and HODII performed electrocorticography recordings and analysis. All authors edited and approved the manuscript.

DECLARATION OF INTERESTS

A patent filed January 14, 2026, is related to the study: Methods and Kits for Achieving Synthetic Nervous System States, PCT/US2026/011285.

The authors declare no other competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES

During the preparation of this work, the authors used AI-assisted technologies [Claude and ChatGPT] for proofreading and grammatical editing. After using these tools, the authors reviewed and edited the content and take full responsibility for the content of the publication.

FIGURE LEGENDS

Figure 1: Single-cell-resolution map of isoflurane-activated brain-wide Fos expression.

(A) Experimental overview: Three complementary pipelines were used to analyze whole brain Fos activation in home cage control and isoflurane-treated mice: voxel-wise immunofluorescence analysis with UNRAVEL (panels D-G), atlas-region-based Fos⁺ puncta quantification with ClearMap (Fig S1), and functional network analysis with SMARTTR (Fig 2). (B) Mean Fos⁺

density shown as a heatmap across the intact brain in all isoflurane samples ($n = 9$). (C) Mean Fos+ density heatmap across all control ($n = 7$, top row) and isoflurane samples ($n = 9$, bottom row), virtually sliced in the coronal plane at major anatomical subdivisions (see [Fig S1](#) for detailed ClearMap analysis and representative Fos images). UNRAVEL mapped regions where isoflurane either (D) suppressed Fos ($q < 0.05$) or (E) enhanced Fos ($q < 0.005$). FDR-corrected clusters differing in Fos+ cell density are shown in mirrored 3D brains and bar graphs (see [Video S1](#) and [Methods](#)). Data are shown as mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, unpaired two-tailed t-tests. (F-G) Sunburst plots summarize the anatomical composition of valid clusters by volume, with outer rings representing finer anatomical granularity in the Allen Mouse Brain Common Coordinate Framework (CCFv3) atlas hierarchy. Abbreviations are defined in [Table S1](#) with these exceptions: {'a': 'anterior', 'd': 'dorsal', 'l': 'lateral', 'v': 'ventral'}. Raw Fos counts and z-scored ClearMap data are provided in [Table S1](#). See [Fig S2](#) and [Table S1](#) for details on each valid cluster and raw cell densities across multiple FDR thresholds. Scale bar, 1 mm.

Figure 2: Network analysis reveals hubs that functionally differentiate an isoflurane unconscious state.

Regional correlation heatmaps for (A) Control and (B) Isoflurane condition based on ClearMap atlas-registered mean Fos density data. Groupings of subregions by their larger parent regions are shown as sub-facets. (C) A combined network (bottom) showing the overlapping functional connectivity patterns of the individual control (top, red) and isoflurane (middle, blue) functional networks with matching network density (edge proportion = 3.15%) and the combined network below (see [Fig S3](#) for proportional thresholding). (D) Functional difference network generated by plotting significant permuted regional correlation differences as edges (lines), and the regions they connect as nodes (circles). The uniquely shaded convex hull polygons denote regions classified as members of the same community. Key community nodes are abbreviated C1: LM = lateral mamillary nucleus; C2: MEPO = median preoptic nucleus; C3: LPB = lateral parabrachial nucleus, C4: LA = lateroanterior hypothalamic nucleus; C5: MiTg = microcellular tegmental nucleus; C6: B = Barrington nucleus; C7: TMv = ventral tuberomammillary nucleus; C8: CEA = central amygdalar nucleus; C9: pEW = pre-Edinger-Westphal nucleus. (E) Anatomical distribution of key community nodes identified in (D) and also shown in [Video S2](#). See [Fig S3](#) and [Table S2](#) for all abbreviations and detailed statistics.

Figure 3. Integration of multimodal behavior and peripheral physiology during isoflurane state transitions.

(A) Mechano-acoustic devices (MA devices) were implanted in mice to obtain internal physiological metrics (heart rate, HR; respiratory rate, RR; temperature & physical activity), integrated with pose estimation and isoflurane (ISO) gas concentration readout for a comprehensive assessment of internal state during isoflurane anesthesia. (B) We obtained physiological measurements as mice ($n = 7$) progressed through isoflurane induction. (C) Behavioral events detected by supervised classifiers using pose estimation. With increasing ISO concentration, mice gradually ceased grooming and rearing behaviors as they exhibited extended immobility. See [Video S3](#) for representative integration of behavior and physiology. (D) Following induction ("ISO onset"), heart rate ("HR"; red), respiratory rate ("RR"; blue), temperature ("Temp."; green) and physical activity (yellow) all decreased with increasing ISO concentration (%), preceded by a brief rise only in HR and physical activity. Horizontal dashed lines show means of baseline trials

(Means $\pm$ SEM; heart rate, 703.0 ± 2.26 bpm; respiratory rate, 223.8 ± 0.56 rpm; body temperature, 38.5 ± 0.01 °C; physical activity, 22.7 ± 0.38 a.u.). See [Table S3](#) for detailed statistics. (E) 3-dimensional plot of pose trajectories reflecting locomotor changes with increasing isoflurane concentration. Color intensity represents ISO concentration from 0 to 1.4%. Scale bar is 10 minutes.

Figure 4: Brain-wide activity-dependent capture of anesthesia unconsciousness.

(A) Experimental timeline: FosTRAP2 transgenic mice received retroorbital injection of AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure (1.2-1.3%) with 4-hydroxytamoxifen (4-OHT); (DREADD = designer receptor engineered to be activated by a designer drug). Mice were tested in the absence of anesthetic after either clozapine-n-oxide (CNO) or saline injection in a modified open field test with concurrent videorecording and peripheral physiology recording using wireless mechano-acoustic (MA) devices, or neural physiology with electrocorticography (ECoG). (B) 1-4 Hz delta slow wave activity is prominent after isoflurane induction and (C) after CNO injection as compared to saline in Gq-DREADD mice ($n = 4$, see [Fig S4B](#) for spectrograms). (D) Gq-DREADD expressing mice fall off a rotarod (4-rpm) faster after CNO injection ($n = 10$). (E) Gq-DREADD mice display loss of righting reflex at a significantly reduced isoflurane concentration after CNO injection, though this does not occur at baseline (see [Fig S4C](#) for dose-response and non-TRAP oil control). (F) A representative raster plot from all 6 (3 saline, 3 CNO) of one animal's trials after analysis of classified behavior (also see [Video S4](#)). (G) Bouts of classified behavior for the average of all saline and CNO trials per animal ($n = 7$ mice, total of 18 CNO and 20 saline trials). (H) Left: Heart rate (beats per minute, BPM) decreases after CNO injection as compared to saline, shown time-aligned to drug onset, right: mean heart rate (HR) decreases after CNO injection in Gq-DREADD mice. (I) Left: Body temperature (Celsius) changes over duration of trial, time-aligned to drug onset, right: mean temperature decreases with CNO. See [Fig S4](#) and [Video S4](#) for Gi-DREADD data. (J) Experimental timeline for anti-nociception testing: FosTRAP2 mice received retroorbital injection of either Cre-dependent activating Gq-DREADD or inhibitory Gi-DREADD. They underwent isoflurane exposure, during which either 4-OHT or oil vehicle (if non-TRAP control) was administered. On test day, mice were tested for anti-nociception following saline injection (Pre) and 5 mg/kg CNO (Post). Warm water tail withdrawal (tail flick) latency was tested at 30-, 60-, and 90-minutes (min) post-CNO injection, and hotplate behavior was tested at 30- and 90-min post-CNO injection interleaved with tail flick tests. (K) Representative hindbrain light sheet image after intact brain iDisco+ clearing and mCherry-immunolabeling for DREADD expression ($n = 7$ mice). Scale bar is 1 mm and 200 μ m in inset. See [Fig 5](#) for brain analysis. (L) Gq-DREADD mice ($n = 8$, orange) show increased latency to tail flick compared to Gi-DREADD ($n = 7$, blue) or non-TRAP control mice ($n = 8$, gray) at every time point tested. (M) In the hot plate test, Gq-DREADD mice do not jump off the hotplate, have fewer paw withdrawals off the hot plate at 90-min, and show increased latency to first paw withdrawal response at 30 and 90-min post-CNO. 2-way repeated measures ANOVA with Tukey's multiple comparisons: * $p < 0.05$, ** $p < 0.01$. Paired t-tests: * $p < 0.05$, ** $p < 0.01$. Error bars are the mean $\pm$ SEM. See [Table S3](#) for detailed statistics.

Figure 5. Spatial distribution of DREADD immunolabeling after brain-wide activity-dependent capture.

(A) Experimental timeline: FosTRAP2 transgenic mice received retroorbital injection of an AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure with 4-hydroxytamoxifen (4-OHT). After experiments, intact brains were immunolabeled for mCherry-DREADD and cleared by iDisco+ ($n = 7$ mice). ClearMap was

used to register brains, detect TRAP+ signal, generate cell counts and density heat maps. **(B)** Mean TRAP+ signal density calculated across larger brain subdivisions after atlas registration by ClearMap shows similar TRAP+ signal density across the brain, except for significant differences in the hindbrain and cerebellum (1-way ANOVA, $*p < 0.05$). **(C)** 3-dimensional mean density heatmap of all samples ($n=7$). **(D)** Mean TRAP+ density across all samples shown as voxel-based 2D-projection heatmaps (calibration bar is arbitrary units). **(E)** Representative whole brain light sheet microscopy images after iDisco clearing and mCherry-DREADD immunolabeling digitally sliced to the 2D coronal plane from one control mouse expressing retroorbital AAV-PHP.eB Cre-dependent Gq-DREADD with oil vehicle administered during the isoflurane exposure (non-TRAP control, see [Methods](#)) and **(F)** one Gq-DREADD-expressing mouse that received 4-OHT during isoflurane exposure for activity-dependent capture of synthetic unconscious state. Abbreviations: BMA = basomedial amygdala, VMH = ventromedial hypothalamus, MiTG = microcellular tegmental nucleus, PrCnF = precuneiform area. Scale bar for full coronal slices is 1mm, inset scale bar is 500 μ m. See [Table S3](#) for detailed counts and statistics.

Figure 6: Brain-wide chemogenetic activation preferentially recruits isoflurane-activated neurons in the parabrachial nucleus.

(A) Experimental schematic. Neuropixels probes (DiI-coated for histological tract reconstruction) were used to record single-unit activity from FosTRAP2 mice expressing Gq-DREADD, across a 5-min awake baseline and isoflurane induction (ISO, 1.3%) maintained for at least 20 min. This was followed by a 25-min wash and second 5-min awake baseline before delivering CNO (5 mg/kg) and monitoring chemogenetic activation of ISO-activated ensembles for 30 min. **(B)** Anatomical localization of recording probe tracts. Left: representative maximum intensity projection (top) across coronal depth (500 μ m) showing DiI probe tract (magenta), which are traced and aligned to anatomical regions in atlas space (bottom), to determine probe trajectories in the brain (right). **(C)** Anatomical distribution of recorded units by major region (color key) and the proportion responsive to CNO (darker bar color) per region; isocortex (Cortex, 235/346 CNO-responsive), hindbrain (244/361), and parabrachial nucleus (PB, 55/95) were targeted for analysis. Total CNO-responsive $N = 1,596/2,076$ total units obtained across 14 recordings, from 8 mice. **(D)** Left: representative firing-rate rasters ($\log_{10}$ FR) for cortex (top), hindbrain (middle) and parabrachial nucleus (bottom) units significantly activated (top, “ $\square$ ”) or suppressed (bottom, “ $\square$ ”) by isoflurane, showing activity during “pre” (3 min leading up to onset) and “post” (3 to 6 min after onset) time periods. Right: distribution of single-unit response magnitudes as a change in their baseline-normalized $\log_{10}$ firing rate (ΔZ , post – pre), as a % of total region population; gray bars denote neutral response distribution. **(E)** CNO-response panel as in **D** for cortex (top), hindbrain (middle) and parabrachial nucleus (bottom). Representative raster plots compare activity 5 min leading up to (“pre”) vs. 15-20 min after (“post”) CNO injection. ISO drove significant responsiveness above empirical null in all regions (CTX 267/346, 77.2%; HB 295/361, 81.7%; PB 66/95, 69.5%; $q < 0.0001$), with responsive units strongly biased toward suppression (of ISO-responders: CTX 248/267, 92.9%; HB 278/295, 94.2%; PB 56/66, 84.8%; $q < 0.0001$). Likewise, CNO engaged a significant proportion in all regions (CTX 235/346, 67.9%; HB 244/361, 67.6%; PB 55/95, 57.9%; all $q < 0.0001$), with a majority being similarly biased toward suppression among CNO-responders (CTX 173/235, 73.6%; HB 180/244, 73.8%; PB 40/55, 72.7%; all $q < 0.0001$). **(F-G)** Percentage of units that were ISO-responsive, computed separately within CNO-responsive (CNO $\square$) and CNO-

nonresponsive (CNO⁻) subgroups (mean $\pm$ Wilson 95% CI). **(F)** In cortex (top), ISO⁺ rates did not differ between CNO⁻ and CNO⁺ units (179/235, 76.2% vs. 88/111, 79.3%; OR = 0.84, $q = 0.82$), whereas in hindbrain (middle), CNO⁺ units overall were significantly enriched in ISO⁺ (209/244, 85.7% vs. 86/117, 73.5%; OR = 2.15, $q = 0.007$). **(G)** The enrichment within hindbrain was significant in LPB (43/55, 78.2% vs. 23/40, 57.5%; OR = 2.65, $q < 0.05$) but not in surrounding non-PB hindbrain (“non-PB-HB”). However, a direct PB vs. non-PB hindbrain comparison (CNO-responsiveness $\times$ region) was in the predicted direction but not significant (Firth-penalized likelihood-ratio test, OR = 1.60, $p = 0.18$). **(H-I)** Upset plots of joint ISO $\times$ CNO classification among double-responsive units, showing the percentage of each direction combination (activated / suppressed). Directional agreement (same sign response across ISO and CNO) did not exceed chance in any region (cortex 70.4%, $q = 0.42$; hindbrain 75.1%, $q = 0.64$; PB 74.4%, $q = 0.32$; nonPB-HB 75.3%, $q = 1$), with same-direction pairs being composed mainly of units suppressed by both. All p -values corrected using Benjamini–Hochberg FDR across 24 tests (6 per region), with significance reported as q values. See [Supplemental Fig S5](#) and [Table S4](#) for detailed statistics and data from additional regions. FR: firing rate; ΔZ : change in baseline-normalized firing rate; OR: odds ratio.

Figure 7: Chemogenetic activity-dependent capture of lateral parabrachial nucleus hub recapitulates components of altered consciousness.

(A) Experimental timeline: FosTRAP2 transgenic mice received bilateral intra-parabrachial injection of AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure (1.2-1.3%) with 4-hydroxytamoxifen (4-OHT); (DREADD = designer receptor engineered to be activated by a designer drug). Mice were tested in the absence of anesthetic after either clozapine-n-oxide (CNO) or saline injection in a modified open field test with concurrent video-recording and peripheral physiology recording using wireless mechano-acoustic (MA) devices. **(B)** Representative image of mCherry-DREADD immunolabeling after viral expression in the LPB. Scale bar is 1 mm; also see [Fig S6](#). **(C)** A representative raster plot from one animal’s classified behavior from the average of CNO and saline trials (also see [Video S5](#)). **(D)** Bouts of classified behavior for the average of all saline and CNO trials per animal ($n = 10$ mice, total of 29 CNO and 47 saline trials). **(E)** Left: Heart rate (beats per minute, BPM) decreases after CNO injection as compared to saline, shown time-aligned to approximated drug onset, right: mean heart rate (HR) decreases after CNO injection in Gq-DREADD mice ($n = 4$). **(F)** Left: Body temperature (Celsius) changes over duration of trial, time-aligned to drug onset, right: mean temperature decreases with CNO ($n = 4$). **(G)** Gq-DREADD mice display loss of righting reflex at a reduced isoflurane concentration after CNO injection ($n = 10$), though this does not occur at baseline. **(H)** Gq-DREADD mice ($n = 10$, brown) show increased latency to tail flick compared to baseline at every time point tested, while Gi-DREADD mice ($n = 7$, green) do not. **(I)** LPB cell-types activated by isoflurane analyzed using Fos maps overlaid with Allen Brain Cell Atlas predominantly consist of glutamatergic cells, as well as GABAergic and cholinergic subpopulations; see [Fig S7](#) for detailed analysis. 2-way repeated measures ANOVA with Sidak’s multiple comparisons: $**p < 0.01$, $***p < 0.001$. Paired t -tests: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. Error bars represent mean $\pm$ SEM. See [Table S5](#) for detailed statistics. See [Fig S6](#) for all Gi-DREADD data.

727 SUPPLEMENTAL FIGURES

728 **Supplemental Figure S1: ClearMap analysis and Raw Fos images show subcortical isoflurane-**
729 **activated hotspots.** (A) Intact whole brains were imaged horizontally with light sheet microscopy to
730 acquire cellular resolution Fos+ signal, then digitally resliced in the coronal plane at the indicated
731 AP positions. (B) Voxel-wise comparison of control vs isoflurane conditions, with cyan indicating
732 higher Fos+ cell densities in the isoflurane condition and red indicating higher Fos+ cell densities in
733 the control condition. (C) Representative mean intensity projections of the raw Fos+ signal (3.6x
734 objective) from the control condition and (D) isoflurane condition virtually re-sliced to the coronal
735 plane at the indicated significant regions, from top to bottom: BNST = bed nucleus of stria
736 terminalis; PVH = paraventricular hypothalamus; CeA = central amygdala; VMH = ventromedial
737 hypothalamus; PB = parabrachial nucleus; LC = locus coeruleus. (E) Voxel-wise significance map
738 of areas significantly regulated in the isoflurane (blue) or control (red) conditions. (F) 3D mean
739 Fos+ density heatmaps from the isoflurane condition (also shown in [Fig 1B-C](#)). (G) Overall mean
740 Fos+ density, calculated across major brain subdivisions after atlas registration by ClearMap in
741 control vs. isoflurane conditions, identifies significant differences in the isocortex, hypothalamus,
742 midbrain and cerebellum (unpaired t-test, * $p < 0.05$). (H) Pairwise analysis of z-scored Fos density
743 by each distinct anatomical subregion after FDR correction ($q < 0.01$) reveals the most significant
744 subregions. Direction of effect in isoflurane relative to control is shown in the bottom bar with
745 green indicating regions of enhanced Fos in isoflurane. Select regions are highlighted (AON:
746 anterior olfactory nucleus, CPu: caudate-putamen, CeA: central amygdala; BNST: bed nucleus of
747 stria terminalis; PVT: paraventricular thalamus; PVH: paraventricular hypothalamus; VMH:
748 ventromedial hypothalamus; SC: superior colliculus; PB: parabrachial nucleus; LC: locus
749 coeruleus). See [Table S1](#) for a full list of regions from pairwise analysis, all raw and z-scored Fos
750 data. $n = 7$ controls, 9 isoflurane.

751
752 **Supplemental Figure S2: Identification of isoflurane-activated clusters from UNRAVEL**
753 **analysis.**

754 Voxel-wise data from [Fig 1D-G](#) were warped to the Allen Mouse Brain Common Coordinate
755 Framework (CCFv3) for display in coronal brain slices and overlaid with a colored wireframe using
756 FSLeaves. Input images for voxel-wise comparisons were averaged for each group to show
757 background-subtracted and z-scored Fos immunofluorescence. FDR-corrected p-value maps show
758 increases (warm colors) and decreases (cool colors) in Fos expression resulting from isoflurane
759 treatment. For each effect direction, the most stringent q-value threshold is fully opaque (100%),
760 the next most stringent is semi-opaque (66%), and the least stringent is less opaque (33%). Valid
761 clusters with significant differences in Fos+ cell density were randomly colored and labeled with
762 their IDs. Abbreviations: Adj. = adjusted; Avg. = average; SS = somatosensory; OT = olfactory
763 tubercle; SI = substantia innominata; BNST = bed nucleus of the stria terminalis; MA =
764 magnocellular nucleus; PVH = paraventricular hypothalamic nucleus; Hypothal. = hypothalamus;
765 CeA = central amygdala; Thal. = thalamus; ZI = zona incerta; COA = cortical amygdalar area; MB
766 = midbrain; Sup. = superior; PAG = periaqueductal gray; Inf. = inferior. Please see [Table S1](#) for a
767 full list of abbreviations and detailed cluster statistics.

768
769
770 **Supplemental Figure S3: Isoflurane networks are denser across a variety of weight thresholds.**

(A) control (red) and isoflurane (blue) networks thresholded at various alpha values. (B) Mean degree, clustering, betweenness, and efficiency metrics calculated across a range of significance thresholds and (C) edge proportion thresholds. (D) Assessment of small-world properties of individual networks plotted in **Fig 2**. Small-world index (σ) for both networks are above the threshold of 1 (black dotted horizontal line). The control network (E) clustering coefficient and (F) path length as well as the isoflurane (G) clustering coefficient and (H) path length are higher than randomly rewired networks with matching degree sequences. (I) Distributions of the Pearson correlation coefficients between all unique regional connections and their associated (J) p-value for the control and isoflurane networks. (K) Volcano plot illustrating the regional correlations which most differ between the networks. The horizontal red line denotes a significance threshold of $p < 0.001$. Red points in the upper right (control < isoflurane) and upper left quadrants (control > isoflurane), defined by vertical red line lines, indicate regional correlation differences with a magnitude shift greater than 1 between groups. Two-tailed Student's t -test: $**p < 0.01$, $****p < 0.0001$. Error bars represent mean $\pm$ SEM. See **Table S2** for full abbreviations and detailed statistics.

Supplemental Figure S4: Electrocorticography and brain-wide chemogenetic behaviors.

(A) Experimental timeline for electrocorticography recordings: FosTRAP2 transgenic mice received retroorbital AAV injection expressing Gq-DREADD and activity-dependent capture during isoflurane exposure, followed by neural electrocorticography (ECoG) recordings over five days (DREADD = designer receptor engineered to be activated by a designer drug, 4-OHT = 4-hydroxytamoxifen, ECoG = electrocorticography, CNO = clozapine n-oxide, ISO = isoflurane). (B) Day 1 (labeled "1 - Pre") shows the average 40-minute baseline recording without manipulations. Recordings on Days 2-4 were time-aligned to the approximated onset of intraperitoneal CNO (10 min post-injection) indicated by the white dotted line. Day 5 (labeled "5 - Post") was obtained without drug manipulation. The spectrograms shown are an average of 4 mice. (C) The concentration of isoflurane (%), as measured with a gas analyzer sampling from the chamber, at which mice display loss of righting reflex during saline or CNO (10 mg/kg and 5 mg/kg) is significantly reduced in Gq-DREADD but not non-TRAP controls that received oil injection instead of 4-OHT ($n = 5-6$ mice). (D) CNO induces dose-dependent reduction in maximum speed (m/s) in the open field in Gq-DREADD mice. (E) CNO increases mean respiratory rate in Gq-DREADD mice. (F) The concentration of isoflurane (%) at which Gi-DREADD mice display loss of righting reflex during saline or CNO (10 mg/kg, 5mg/kg) is unchanged. (G) Maximum speed (m/s) is unchanged in Gi-DREADD mice. (H) Representative raster plot from all 6 (3 saline, 3 CNO) of one Gi-DREADD animal's trials after analysis of classified behavior. (I) Bouts of classified behavior for the average of saline and CNO (5 mg/kg) trials per animal expressing inhibiting Gi-DREADD virus ($n = 5$ mice). There is no change in (J) heart rate (BPM, beats per minute) (K) respiratory rate (breaths per minute) and (L) temperature (Celsius) as recorded from the wireless mechano-acoustic device during trials shown in H-I ($n = 6$ mice, 2-3 trials per drug condition). (M) Rotarod (4-rpm) performance in Gi-DREADD mice after 5 mg/kg CNO ($n = 5$ mice) is unchanged. Error bars show mean $\pm$ SEM. 2-way repeated measures ANOVA with Tukey's post-hoc tests or paired t -tests, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. See **Figure 4** and **Methods** for experimental details. See **Table S3** for detailed statistics.

Supplemental Figure S5: Brain-wide TRAP Neuropixels Supplemental Data.

(A) ISO-response panels and (B) CNO-response panels for regions not included for analysis in Fig 6. Left: mean baseline-normalized log firing rate traces (“z-score”) for units significantly activated (top, lighter) or suppressed (bottom, darker) by isoflurane, shown pre vs. post onset. Right: distribution of single-unit response magnitudes as a change in their baseline-normalized firing rate (ΔZ , post – pre), as a % of total region population; gray bars denote neutral response distribution. (C) Percentage of units that were ISO-responsive, computed separately within CNO-responsive (CNO+) and CNO-nonresponsive (CNO-) subgroups (mean $\pm$ Wilson 95% CI). (D) Upset plots of joint ISO $\times$ CNO classification among double-responsive units, showing the percentage of each direction combination (activated / suppressed). (E) Representative maximum intensity projection (MIP) across coronal depth (400 μ m) showing mCherry-DREADD expression (red) and DiI probe tract (cyan) in parabrachial nucleus (inner dashed outline). (F-G) Forest plots illustrating CI and detailed statistics for (F) proportion tests against region and global empirical shuffle nulls and (G) conditional proportion tests for odds ratio of ISO+ given CNO+ / CNO- (log scale; dashed line at OR = 1, equal odds). See also Table S4 for detailed counts and statistics. FR: firing rate; OR: odds ratio.

Supplemental Figure S6: Effects of chemogenetic inhibition after activity-dependent capture of lateral parabrachial nucleus hub.

(A) Latency to fall (s) in Gq-DREADD expressing mice is unchanged following CNO injection (5 mg/kg, n = 10). (B) Maximum speed (m/s) is significantly reduced in Gq-DREADD mice 15-20 min post-CNO (5 mg/kg, n = 10). There is no change in (C) loss of righting reflex (% isoflurane), (D) latency to fall (s) and (E) maximum speed (m/s) in Gi-DREADD mice following CNO injection (5 mg/kg, n = 7). (F) A representative raster plot from one animal after analysis of classified behavior (also see Video S5) after CNO injection as compared to saline in Gi-DREADD mice. (G) Bouts of classified behavior for the average of saline and CNO trials per animal (n = 7, total of 21 CNO and 31 saline trials) expressing inhibiting Gi-DREADD virus. CNO had no effect on rearing, grooming, freezing, or circling in Gi-DREADD mice. (H) Respiratory rate in Gq-DREADD mice is unchanged after CNO injection as compared to saline (n = 4). (I) Heart rate (BPM, beats per minute) is unchanged following CNO in Gi-DREADD mice (n = 3). (J) Respiratory rate is decreased following CNO in Gi-DREADD mice (n = 3). (K) Temperature (Celsius) is unchanged following CNO in Gi-DREADD mice (n = 3). (L) Representative image after mCherry-immunolabeling for DREADD expression and DAPI staining. Scale bar is 1 mm and 200 μ m in inset. (M) Representative brain sections showing DREADD expression in Gq- (top) and Gi- (bottom) DREADD expressing mice. Highlighted regions indicate areas of viral expression. See Figure 7 and Methods for experimental details. Error bars show mean $\pm$ SEM. Mixed effects analysis with Sidak’s multiple comparisons (panel b, e) and paired t-tests, *p<0.05, ****p<0.0001. See Table S5 for statistics.

Supplemental Figure S7: Diversity of neurons in LPB cluster 3.

(A) Neuron types from the Allen Brain Cell Atlas (ABCA) within the LPB portion of Control < Isoflurane cluster 3 (from Fig. 1D-G) are highlighted in MERFISH-CCF space. For this, c-Fos

clusters were warped to ABCA MERFISH-CCF space. The spatial distributions of *Atoh1*- and *Lmx1*-derived LPB clades, described by Pauli et al. (2022)⁵¹, are noted for reference. **(B)** ABCA neuron type proportions in the LPB portion of cluster 3. **(C)** To examine the alignment of cluster 3 with LPB subregions, the Unified Mouse Brain Atlas v2 labels were warped to MERFISH-CCF space. LPBE, external LPB; LPBD, dorsal LPB. Imputed Calca expression in ABCA neurons is shown in the context of LPB subregions and cluster 3; display range, 0–6 log₂ (CPM + 1). **(D)** Calca marks CGRP-associated LPB neurons, including N15 and N16 subclusters described by Pauli et al. (2022). **(E)** Major LPB projection pathways adapted from Pauli et al. (2022). *Atoh1*-related LPBD populations project predominantly through the ventral pathway, whereas *Lmx1*-related LPBE populations project predominantly through the central tegmental tract (CTT). Both have additional descending projections. Clade proportions were estimated using imputed marker expression, with marker-positive neurons defined as >3 log₂ (CPM + 1): *Atoh1* = *Slc17a6*⁺/*Meis2*⁺/*Lmx1b*⁻, *Lmx1* = *Slc17a6*⁺/*Meis2*⁻/*Lmx1b*⁺, inhibitory = GABAergic neurons, and other = remaining neurons. Target labels indicate c-Fos upregulation in the Control < Isoflurane UNRAVEL analysis: bold, FDR q < 0.005; regular black, 0.005 ≤ q < 0.05; gray, not detected at q < 0.05. Nearly all LPB target regions shown had increased c-Fos, suggesting that isoflurane-activated glutamatergic LPB neurons may contribute to widespread downstream activity. **(F)** Imputed marker expression across ABCA neurons within the LPB portion of cluster 3. *Slc17a6* encodes VGLUT2, *Slc32a1* encodes VGAT, *Chat* encodes ChAT, and *Slc6a5* encodes GlyT2. The transcription factors *Lmx1a* and *Lmx1b* are expressed in the *Lmx1* clade, whereas *Evx2* and *Meis2* are expressed in the *Atoh1* clade. Examples of neuropeptides preferentially associated with *Atoh1*- or *Lmx1*-related LPB populations are shown. Region abbreviations in **(E)**: AI, agranular insula; BNST, bed nucleus of the stria terminalis; CeA, central amygdala; CM, central medial thalamus; DMH, dorsomedial hypothalamus; DR, dorsal raphe; IMD, intermediodorsal thalamus; LHA, lateral hypothalamic area; LS, lateral septum; MEPO, median preoptic nucleus; NTS, nucleus of the solitary tract; PAG, periaqueductal gray; PSTN, paraventricular nucleus; PVH, paraventricular hypothalamus; RE, nucleus of reuniens; RN, reticular nucleus; SI, substantia innominata; VMH, ventromedial hypothalamus; VPMpc, ventral posteromedial thalamus, parvocellular; VTA, ventral tegmental area.

SUPPLEMENTAL TABLES

Supplemental Table S1: Brain-wide Fos detailed statistics and supplemental data (Fig 1)

Supplemental Table S2: Network analysis permutation results, unique regions analyzed, and detailed network community node centrality statistics (Fig 2)

Supplemental Table S3: Brain-wide TRAP Supplemental Data: Mechano-acoustic internal state physiology during isoflurane (Fig 3), extended behavior and electrocorticography results (Fig 4), behavioral classifier statistics (Fig 4), Brain-wide TRAP ClearMap data (Fig 5)

Supplemental Table S4: Neuropixels high density single unit recordings detailed statistics and anatomical distributions (Fig 6)

Supplemental Table S5: Parabrachial TRAP detailed behavior, Parabrachial MERFISH data (Fig 7)

903

904 SUPPLEMENTAL VIDEOS

905 **Video S1.** Anesthesia-activated Fos clusters and their spatial relationship to Neuropixels
906 recordings. Rotating 3D brain models show UNRAVEL clusters representing increases or decreases
907 in Fos⁺ cell density after isoflurane. Subsequent views overlay Neuropixels units and recording
908 channels, including units within the LPB Fos cluster.

909 **Video S2.** Anatomical distribution of community network analysis video. Rotating 3D brains depict
910 key community nodes from functional network analysis.

911 **Video S3.** Isoflurane induction with dynamic mechanoacoustic physiology recording and
912 automated behavioral classification (see [Fig 3](#)).

913 **Video S4.** Brain-wide isoflurane-activated ensemble chemogenetic induction with mechanoacoustic
914 physiology recording and automated behavioral classification. Representative video clips at 4X
915 speed from one Gq-DREADD mouse and one Gi-DREADD mouse's saline and CNO trials after
916 brain-wide TRAP under isoflurane (see [Fig 4](#)).

917 **Video S5.** Lateral parabrachial ensemble chemogenetic induction with dynamic mechanoacoustic
918 physiology recording and automated behavioral classification. Representative video clips at 4X
919 speed from one Gq-DREADD mouse and one Gi-DREADD mouse's saline and CNO trials after
920 LPB TRAP (see [Fig 7](#)).

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1203 **STAR[®]METHODS**

1204 **KEY RESOURCES TABLE**

1205

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rat anti-cFos	Synaptic Systems	RRID: AB_2864765; Cat# 226 017
Rabbit anti-RFP	Rockland	RRID: AB_2209751; Cat# 600-401-379
Donkey anti-Rat AlexaFluor 647 Fab2-conjugated secondary antibody	Jackson ImmunoResearch Labs	RRID: AB_2340696; Cat# 712-606-153
Donkey anti-Rabbit AlexaFluor 647 Fab2-conjugated secondary antibody	Jackson ImmunoResearch Labs	RRID: AB_2340625; Cat# 711-606-152
Bacterial and virus strains		
AAV-PHP.eB-DIO-Gq-DREADD-mCherry	Addgene	Cat# 44361-PHPeb
AAV-PHP.eB-DIO-Gi-DREADD-mCherry	Addgene	Cat# 44362-PHPeb

Biological samples		
Allen Brain Cell Atlas	Allen Institute	https://portal.brain-map.org/atlas-and-data/bkp/abc-atlas
Unified Brain Atlas	Chon et al., 2019 ¹	https://figshare.com/articles/dataset/Unified_mouse_brain_atlas_v2/25750983
Allen Mouse Brain Atlas	Allen Institute	RRID:SCR_002978; https://mouse.brain-map.org
Chemicals, peptides, and recombinant proteins		
Clozapine-N-Oxide	Enzo Life Sciences	Cat# BML-NS105-0025
Deschloroclozapine	Sigma Aldrich	Cat# SML3651
4-hydroxytamoxifen	Sigma Aldrich	Cat# H6278
DiI fluorescent dye	Invitrogen	Cat# V22888
Sesame oil	Sigma Aldrich	Cat# S3547
Critical commercial assays		
None		
Deposited data		
Data included in supplemental tables		
Experimental models: Cell lines		
None		
Experimental models: Organisms/strains		
Mouse: C57bl/6J	Jackson Labs/bred in-house	RRID:IMSR_JAX:000664; Strain# 000664

Mouse: Fos ^{2A-iCreER} (TRAP2)	Jackson Labs/bred in-house	RRID:IMSR_JAX:030323; Strain# 030323
Software and algorithms		
ImageJ	Schneider et al., 2012 ²	RRID:SCR_003070; https://imagej.nih.gov/ij/
SLEAP	Pereira et al., 2022 ³	RRID:SCR_021382; https://sleap.ai/
SimBA	Goodwin et al., 2024 ⁴	RRID:SCR_021413; https://github.com/sgoldenlab/simba
SMARTTR	Jin et al., 2025 ⁵	https://mj1812.github.io/SMARTTR/
UNRAVEL	Rijsketic et al., 2023 ⁶	https://github.com/b-heifets/UNRAVEL
ClearMap	Renier et al., 2016 ⁷	https://github.com/ClearAnatomics/ClearMap
Ilastik	Berg et al., 2019 ⁸	RRID:SCR_015246; https://www.ilastik.org/
Arivis Pro	Zeiss	RRID:SCR_018000; https://www.zeiss.com/microscopy/us/products/software/arivis-pro.html
GraphPad Prism	GraphPad Software	RRID:SCR_002798; https://www.graphpad.com/
BioRender	BioRender	RRID:SCR_018361; https://www.biorender.com/
Any-MAZE	Stoelting Co.	RRID:SCR_014289; https://stoeltingco.com/Neuroscience/ANY-maze
ANTsPy	ANTsX consortium	RRID:SCR_004757; https://github.com/ANTsX/ANTsPy
FSL (FSLeyes, randomise_parallel)	FMRIB, Oxford	RRID:SCR_002823;

		https://fsl.fmrib.ox.ac.uk/fsl
SpikeGLX	Janelia Research Campus, VA	http://billkarsh.github.io/SpikeGLX/
Kilosort4	Pachitariu et al., 2024 ⁹	RRID:SCR_016422; https://github.com/MouseLand/Kilosort
Lasagna	Campbell et al., 2020 ¹⁰	https://zenodo.org/records/3941894
Brainrender	Tyson et al., 2021 ¹¹	RRID:SCR_022328; https://github.com/brainglobe/brainrender

1206

1207 **METHOD DETAILS**

1208 *Mice.* All experiments were approved by the National Institutes of Health and the University of
1209 Washington Institutional Animal Care and Use Committee. 3- to 6-month-old male and female
1210 C57bl/6J or homozygous Fos^{2A-iCreER} (FosTRAP2) transgenic mice were used in all studies.
1211 Animals were group-housed in a 12-hour reverse light/dark cycle facility with *ad libitum* access to
1212 food and water. Animals used in electrocorticography or Neuropixels studies were singly housed
1213 during the recording experiments.

1214

1215 *Intact brain tissue collection after isoflurane exposure:* Mice were singly housed in the home cage
1216 for 3 days prior to tissue collection for intact whole brain Fos immunolabeling. Habituation to the
1217 testing room was achieved by placing mice in a temperature controlled red-light room for at least 2-
1218 3 hours a day for 3 days prior to tissue collection. On the test day, mice in control condition
1219 remained singly housed in their home cage inside the testing room where isoflurane was delivered.
1220 Mice in the isoflurane condition were individually induced with 2% isoflurane and after loss of toe
1221 pinch reflex was confirmed, each mouse was maintained under general anesthesia for 180 minutes

1222 with 1.2%-1.3% isoflurane in oxygen delivered via nose cone. The concentration of isoflurane
 1223 delivered at each nose cone was confirmed to be within range using a gas analyzer prior to placing
 1224 the mouse in the nose cone. Mice were actively warmed on a heating pad and continuously
 1225 monitored by the experimenter. After 180 minutes, all mice were more deeply anesthetized with
 1226 isoflurane and transcardially perfused with phosphate-buffered saline (PBS) followed by 10%
 1227 formalin. Brains were dissected and post-fixed in formalin for 24 hours at 4°C prior to intact whole
 1228 brain clearing and immunolabeling.

1229

1230 *Intact whole brain clearing and immunolabeling:* A previously published, modified version of the
 1231 iDisco+ protocol was used to immunolabel and clear intact brain samples^{7,12}. The following
 1232 antibodies were used: Rat anti-cFos (1:2000; Synaptic Systems) for Fos immunolabeling or Rabbit
 1233 anti-RFP (1:2000; Rockland) for mCherry DREADD immunolabeling, followed by Donkey anti-
 1234 Rat or Donkey anti-Rabbit AlexaFluor 647 Fab2-conjugated secondary antibody (1:500; Jackson
 1235 ImmunoResearch).

1236

1237 *Intact whole brain imaging:* Cleared, intact whole brains were imaged with a uniform axial
 1238 resolution light sheet microscope (SmartSPIM, LifeCanvas Technologies) using either the 1.6x (5
 1239 µm voxels) or 3.6x (1.8 µm voxel) objectives for Fos samples, and 3.6x for TRAP samples.
 1240 Samples were mounted horizontally using a custom sample holder immersed in dibenzyl ether
 1241 (DBE). Images were acquired in two channels (488 nm: autofluorescence; 639 nm:
 1242 mCherry/AlexaFluor-647) and stitched using the SmartSPIM acquisition software. TRAP+ images
 1243 were down-sampled (0.5x) prior to image processing with isotropic 3.6 µm voxels. 3-dimensional
 1244 images were visualized using Arivis Pro software (Zeiss).

1245

1246 *Retroorbital virus injection:* FosTRAP2 mice were briefly anesthetized with isoflurane (5-10-min)
1247 and administered adeno-associated virus (AAV) by retroorbital injection to one eye. AAV-PHP.eB-
1248 hSyn-DIO-hM3D(Gq)-mCherry (44361-PHPeB, Addgene) was used for activating DREADD and
1249 AAV-PHP.eB-hSyn-DIO-hM4D(Gi)-mCherry (44362-PHPeB, Addgene) was used for inhibitory
1250 DREADD. Virus was diluted in total 100μL sterile saline to $\sim 2\text{-}3 \times 10^{11}$ GC/mL based on prior
1251 studies¹³.

1252

1253 *Lateral parabrachial nucleus viral injection:* FosTRAP2 mice were anesthetized with isoflurane
1254 (1.3%-3%) and head-fixed for survival surgery. They also received single injections of lidocaine-
1255 bupivacaine and ketoprofen for peri-operative analgesia. Stereotactic coordinates were standardized
1256 relative to Bregma and Lambda distance, and a Hamilton syringe was used to inject 0.4ml of virus
1257 at a rate of 0.2ml/min. AAV-PHP.eB-hSyn-DIO-hM3D(Gq)-mCherry (44361-PHPeB, Addgene) or
1258 AAV-PHP.eB-hSyn-DIO-hM4D(Gi)-mCherry (44362-PHPeB, Addgene) was injected bilaterally
1259 into the PBN (AP: -4.90mm, ML: +/-1.35mm, DV: -3.40mm). Mice recovered from surgery for at
1260 least two weeks prior to TRAP, followed by at least 1-week post-TRAP recovery prior to
1261 behavioral testing.

1262

1263 *Activity-dependent labeling (TRAP):* 3-4 weeks following retroorbital or intracranial virus injection,
1264 FosTRAP2 mice were induced with 2% isoflurane and then maintained unresponsive at 1.2-1.3%
1265 for a total time of 180 min. Mice were actively warmed on a heating pad and continuously
1266 monitored by an experimenter. A single 50 mg/kg intraperitoneal (i.p.) injection of 4-
1267 hydroxytamoxifen (4-OHT, Sigma Aldrich), was administered at the 90-min halfway point during

1268 the isoflurane exposure to induce *Cre*-recombination and activity-dependent labeling¹⁴. Non-TRAP
1269 control mice also received retroorbital virus injections and isoflurane exposure but then sesame oil
1270 vehicle (Sigma Aldrich) in the equivalent volume of 4-OHT (~0.2mL) was given i.p. at the 90-min
1271 halfway point.

1272

1273 *Clozapine-n-oxide administration*: Clozapine-n-oxide (CNO, Enzo Life Sciences) was solubilized
1274 in water and stored as 5mg/mL aliquots. For testing, the aliquot was diluted in sterile saline to an
1275 end concentration of 0.5 mg/mL for 5 mg/kg behavioral studies. Mice received i.p. injections
1276 according to their body weight ranging from 0.2-0.3 mL for a 20-30g mouse. The equivalent
1277 volume i.p. sterile saline was delivered for saline trials.

1278

1279 *Deschloroclozapine administration*: Deschloroclozapine (DCZ, Millipore Sigma) was solubilized in
1280 DMSO and stored as 2mg/mL aliquots. For testing, the aliquot was diluted in sterile saline to an end
1281 concentration of 0.01mg/mL for 0.1mg/kg behavioral studies. Mice received i.p. injections of DCZ
1282 according to their body weight.

1283

1284 *ClearMap brain-wide TRAP+ analysis*: We processed 7 intact formalin-fixed brain samples from
1285 FosTRAP2 transgenic mice infected with AAV-PHP.eB-DIO-Gq-DREADD-mCherry virus (see
1286 [Retroorbital virus injection](#) and [Activity-dependent labeling](#) procedures) using a modified version
1287 of the ClearMap pipeline as previously published^{12,15}. We trained a pixel classifier in Ilastik⁸ to
1288 segment and quantify single cells based on somatic signal classification. This was achieved by
1289 selecting 21 image tiles (200 × 200 × 220 μm) from three separate samples, cropped from dorsal
1290 (2), central (3), and ventral (2) positions of the brain. To improve segmentation accuracy, an

1291 additional three tiles were used to correct for artifacts in ventricular and boundary regions. The fully
1292 trained classifier was then applied to all samples, generating segmented cell counts per brain region
1293 and voxelized heatmaps. We computed TRAP+ cell counts within two different levels of hierarchy
1294 within the Unified Brain Atlas¹: 1) 11 major anatomical divisions ([Fig 5](#)) and 914 minor subregions
1295 within those high-level anatomical divisions that no longer split into further daughter regions. We
1296 provide the raw TRAP+ cell counts, cell density, and statistical results for all analyses in [Table S3](#).

1297

1298 *ClearMap brain-wide Fos analysis:* We processed 16 intact brain samples (n = 7 controls and n = 9
1299 isoflurane) using a modified version of the ClearMap pipeline as previously published^{7,12}. We
1300 computed Fos-positive activated cell counts within two different levels of hierarchy within the
1301 Unified Brain Atlas¹: 1) 11 major anatomical divisions and 914 minor subregions within those high-
1302 level anatomical divisions that no longer split into further daughter regions ([Fig 1A-C](#); [Fig S1](#)). To
1303 ensure accuracy within the hierarchical relationships, we ensured there were no overlapping spatial
1304 footprints and no double counting of any regions, following each subregion branch until the end or
1305 “stop-level.” We transformed the raw Fos cell counts to z-scores and normalized the data to account
1306 for regional volume differences. We used the home cage control group (control) to normalize the z-
1307 scores for each region of interest using the formula $z = (x - \mu) / \sigma$, where x represents the Fos count of
1308 the treatment group, μ represents the mean Fos count of the control group, and σ represents the SD
1309 of the control group. We then converted the z-score into -log10 for visualization. For the high-level
1310 analysis, we used an unpaired t-test, *p < 0.05 ([Fig S1G](#)). For the stop-level analysis, we used a t-
1311 test followed by the false discovery rate (FDR) multiple comparison correction with an alpha
1312 significance level of 0.01 (q < 0.01), two-tailed ([Fig S1H](#)). We provide the raw Fos counts, z-
1313 scored data and statistical results for all analyses in [Table S1](#).

1314

1315 *UNRAVEL Fos analysis*: We also analyzed whole brain Fos immunolabeling using UNRAVEL⁶
 1316 (**Fig 1D-G, Fig S2, Fig S7; Video S1**) (UN-biased high-Resolution Analysis and Validation of
 1317 Ensembles using Light sheet images), with refactored Python code (documentation: b-heifets.github.io/UNRAVEL/index.html). Autofluorescence images were resampled to 50 μm
 1318 resolution, and tissue was masked using Ilastik (v1.4.0; pixel classification; all features) for N4 bias
 1319 field correction (ANTsPy; 0.4.2). The resampled images were padded (15% voxels on all sides) and
 1320 smoothed with a Gaussian filter (sigma = 0.4) before being registered with an average template
 1321 from iDisco and light-sheet microscopy (LSFM), which was aligned with the Allen Mouse Brain
 1322 Common Coordinate Framework (CCFv3)¹⁶. Registration was performed using ANTsPy:
 1323 `ants.affine_initializer(fixed_image=autofluorescence_image, moving_image=template,`
 1324 `search_factor=1, radian_fraction=1, local_search_iterations=500),` followed by
 1325 `ants.registration(fixed=autofluorescence_image, moving=template_aligned_with_tissue,`
 1326 `grad_step=0.1, syn_metric='CC', syn_sampling=2, reg_iterations=(100, 70, 50, 20)).` Registration
 1327 accuracy was visually confirmed using FSLeves (v0.30.1; FMRIB), with the atlas warped to each
 1328 fixed registration input image. Fos immunofluorescence was enhanced through 3x3x3 spatial
 1329 averaging and rolling ball background subtraction (pixel radius = 4). The resulting images were
 1330 warped to 25 μm atlas space using transforms from registration. Each hemisphere was z-scored
 1331 using a warped tissue mask and a hemispheric mask ((image – mean intensity in the brain)/standard
 1332 deviation of intensity in the brain). The images were smoothed with a 100 μm kernel and mirrored
 1333 to average the left and right sides. A unilateral atlas mask, excluding ventricles, fiber tracts, and
 1334 undefined regions, spatially restricted the voxel-wise analyses.

Voxel-wise comparisons between the isoflurane and control groups (t-test design) were performed using non-parametric permutation testing (18,000 permutations) with the general linear model (randomise_parallel; FMRIB; v6.0.2). P-value maps were corrected for multiple comparisons using the false discovery rate (FDR) method, applied across a series of q-values to identify clusters of significant voxels. Data in **Fig 1D-G** are from the most stringent q-value thresholds ($q < 0.005$, $p < 0.000056$ for Control < Isoflurane; $q < 0.05$, $p < 0.0012$ for Control > Isoflurane). Clusters smaller than 100 voxels were excluded. Clusters were warped back to full-resolution tissue space for cell density measurements (Fos+ cells per cluster volume). Fos+ cells were segmented using Ilastik. 3D counting was performed using connected components-3d (v3.16.0; connectivity = 6). Clusters were considered valid if unpaired t-tests confirmed that differences in Fos labeling reflected changes in Fos+ cell densities. Information on valid clusters, including regional composition and Fos+ cell densities across FDR q-values, is provided in **Table S2**. Representative images are provided in **Fig S2**. Statistical maps and other related data are available upon request. Scripts from [UNRAVEL](https://github.com/b-heifets/UNRAVEL) (github.com/b-heifets/UNRAVEL) are available at zenodo.org/records/13655988.

UNRAVEL LPB cell types and marker expression analysis: To characterize the cellular composition of the isoflurane-activated lateral parabrachial nucleus (LPB) ensemble, the Control < Isoflurane UNRAVEL cluster 3 mask from the whole-brain Fos analysis was transformed into Allen Brain Cell Atlas (ABCA) MERFISH-CCF space using UNRAVEL. The LPB portion of cluster 3 was defined by intersecting the transformed cluster mask with the PB label in MERFISH-CCF space. Unified Mouse Brain Atlas v2 labels were also transformed into MERFISH-CCF space to visualize the relationship between cluster 3 and LPB subregions, including external LPB (LPBE) and dorsal

1359 LPB (LPBD). Neurons located within the LPB portion of cluster 3 were retained for downstream
1360 analysis and summarized using ABCA neuronal taxonomy annotations.

1361 Cell-type composition was summarized across the ABCA hierarchy, including
1362 neurotransmitter class, class, subclass, supertype, and cluster. For visualization in [Fig S7](#), sunburst
1363 plots were collapsed to three ABCA levels for readability, whereas [Table S5](#) provides summaries
1364 across all five ABCA annotation levels. In sunburst plots, wedge size represents the percentage of
1365 LPB cluster 3 neurons assigned to each ABCA category. For marker-expression plots, wedge color
1366 represents mean imputed expression, shown as $\log_{10}(\text{CPM} + 1)$.

1367 Using imputed MERFISH data from Yao et al. (2023)¹⁷ for each retained neuron, marker
1368 gene expression was quantified for all present cell types. Marker selection was guided by the LPB
1369 molecular and projection framework described by Pauli et al. (2022)¹⁸. For each marker gene and
1370 ABCA annotation level, we calculated mean imputed expression and the percentage of marker-
1371 positive neurons. Marker-positive neurons were defined as cells with imputed expression >3
1372 $\log_{10}(\text{CPM} + 1)$. *Atoh1*-related glutamatergic neurons were operationally defined as
1373 *Slc17a6+/Meis2+/Lmx1b-*, *Lmx1*-related glutamatergic neurons as *Slc17a6+/Meis2-/Lmx1b+*,
1374 inhibitory neurons as GABAergic neurons, and remaining neurons as other.

1375 For the [Table S5](#) overview, each marker was summarized by identifying, at each ABCA
1376 annotation level, the most prevalent category within LPB cluster 3 among categories with mean
1377 imputed expression $>3 \log_{10}(\text{CPM} + 1)$. Percent values in the ‘Marker overview’ tab indicate the
1378 percentage of LPB cluster 3 neurons assigned to the listed ABCA category. Categories with no
1379 mean expression above threshold at a given ABCA level were denoted as not detected (n.d.).
1380 Proportions of all present cells can be found in the ‘LPB cluster 3 neurons’ tab. Look up tables in

1381 the ‘LUT’s’ tab can apply sunburst plot coloring in Flourish. Other tabs characterize marker gene
1382 expression across cell types.

1383 Major LPB projection pathways and target regions were adapted from Pauli et al. (2022)¹⁸.
1384 To relate LPB projection targets to isoflurane-induced Fos activity, mapped target regions were
1385 annotated using the Control < Isoflurane UNRAVEL analysis, noting Fos upregulation in LPB
1386 target regions at FDR $q < 0.005$ and $0.005 \leq q < 0.05$.

1387

1388 *Correlational, Permutation, and Network Analysis:* All associated network analyses and
1389 visualizations were conducted with the SMARTTR⁵ package, igraph, tidygraph package, and
1390 custom written functions (Fig 2). Pearson correlations between normalized Fos⁺ activity per region
1391 were calculated and asymptotic p-values for each correlation were determined using a one-sample t-
1392 test. Significantly correlated regions are visualized in heatmaps after FDR (Benjamini-Hochberg)
1393 correction using a significance threshold of $q < 0.05$. A permutation analysis was conducted to
1394 identify the functional connections that differed most between experimental groups. Correlation
1395 differences were calculated by subtracting correlations in the Control group from respective
1396 correlations in the Isoflurane group ($r_{\text{Isoflurane}} - r_{\text{Control}}$). Each correlation difference was used as a
1397 test statistic against a null distribution produced by shuffling the labels of Control and Isoflurane
1398 mice 10,000 times, with the correlation difference recomputed for each shuffle. Each test statistic
1399 was compared to its individual null distributions to determine the p-value. Significance correlation
1400 differences for all analyses were thresholded using an alpha value of 0.001 and visualized as a
1401 constructed network. Edge (line) thickness in the permuted network represents the magnitude of
1402 correlation difference, with regions represented as nodes. An agglomerative fast greedy modularity
1403 optimization algorithm¹⁹ was applied using tidygraph for community detection.

Thresholding is normally applied to assess the topological features of the most salient connections, although there is little consensus on the best approach, (i.e., thresholding by connection strength or connection proportion) or choice of threshold²⁰. Thus, we constructed networks for Control and Isoflurane across a range of alpha thresholds (FDR uncorrected) from 0.0001 to 1, and across a range of edge proportion thresholds, from 0.001 to 0.50.

In both the permuted and individual functional networks, several topology metrics per node, such as degree, clustering coefficient, efficiency, and betweenness centrality²¹ are automatically calculated using the `create_networks()` or `create_permutation_network()` functions in SMARTTR. Summary statistics are calculated using the `summarise_networks()` function, which averages nodal metrics to calculate mean degree, mean efficiency, mean clustering coefficient, and mean betweenness for a given network. To assess for “small-world” properties²² of empirical Control and Isoflurane networks (edge proportion of 3.15%), we compared the clustering coefficient (C) and characteristic path length (L) with the corresponding values (C^{rand} , L^{rand}) averaged from 100 random null networks. Null networks are generated by applying the degree sequence-preserving maslov-sneppen²³ rewiring algorithm with the `rewire_network()` function, and averaged metrics were calculated with `summarize_null_networks()`. Number of edge rewirings per random network are equivalent to number of edges x 100. The small-world index (s), a ratio summarizing the propensity for high clustering while maintaining efficiency in a network, is defined here as [Equation]. Values above 1 are considered indicative of small-world properties. All analysis code is available on (<https://mj1812.github.io/SMARTTR>). See also **Tables S3-S5** for detailed statistics.

1426 *Loss of righting reflex:* Latency to loss of righting reflex was assessed in a chamber equipped for
 1427 isoflurane delivery and sampling of isoflurane concentration with a gas analyzer ([Fig 4](#), [Fig 7](#), [Fig](#)
 1428 [S4](#) and [Fig S6](#)). Mice were individually injected 5mg/kg CNO, 10mg/kg CNO, 0.1mg/kg DCZ, or
 1429 saline with 10 mg/kg CNO or saline in alternating trials at least one day apart and placed in the
 1430 chamber 15 minutes after injection. Isoflurane was administered stepwise from 0.5 to 2% on the
 1431 vaporizer in 0.5 increments and loss of righting reflex was assessed at each stepwise increment with
 1432 the exact sampled concentration recorded.

1433

1434 *Warm water tail withdrawal test:* Mice were tested for the latency to remove their tail from a warm
 1435 water bath (52.5°C). Mice are restrained within a paper towel, and 1/3 to 1/2 of the tail is immersed,
 1436 and the time to remove the tail (tail flick latency) is recorded^{24–27}. This is done at baseline after
 1437 saline, and then in 30-min intervals after CNO injection. See [Table S3](#) and [Table S5](#) for detailed
 1438 statistics.

1439

1440 *Hot plate test:* Mice are placed on a 53°C plate for 30 seconds, and behaviors are video recorded²⁸.
 1441 This occurs at baseline and after CNO at 2-time intervals (30-min and 90-min post-CNO)
 1442 interleaved between tail withdrawal tests. Latency to jump, groom paws, and number of
 1443 occurrences for these behaviors are quantified from videos by two independent researchers. See
 1444 [Table S3](#) for statistics.

1445

1446 *Rotarod test:* Coordination post CNO injection was tested for each mouse on a rotarod²⁹ after
 1447 modified open field testing. Mice were habituated for two days, 10-min per day, at 4 rotations per
 1448 minute (rpm) before testing. On test day, mice received a saline i.p. injection in the morning and

1449 were placed on the rotarod at 4 rpm. Latency to fall was recorded, unless animals stayed on for the
1450 duration of the experiment (10 min). Mice were recorded again in the afternoon with a 5mg/kg i.p.
1451 CNO injection. See [Table S3](#) and [Table S5](#) for statistics.

1452

1453 *Annotation:* Behavior was annotated in 20-min video clips obtained from the modified open field
1454 test. For fitting the Straub tail classifier, we used a sampled annotated dataset where the behavior
1455 was present in 24244 frames (808s) and behavior was absent in 627312 frames (20910s). For fitting
1456 the grooming classifier, we used a sampled annotated dataset where the behavior was present in
1457 47005 frames (1566s) and behavior absent in 94010 frames (3133s). For fitting the rearing
1458 classifier, we used a sampled annotated dataset where the behavior was present in 31596 frames
1459 (1053s) and behavior absent in 315960 frames (10530s).

1460

1461 *Featurization:* We featurized video and pose-estimation data using image, movement, geometry,
1462 circular- and frequentist statistical methods (see below). For acceptable runtimes, we used GPU
1463 accelerated and multiprocessing methods available through the SimBA API⁴. Features were
1464 computed in metric space after calculating the pixel-to-metric conversion factors in the SimBA
1465 graphical interface. Classifications were smoothened to remove any detected bouts less than 200ms
1466 in length. Computations are primarily dependent on numba³⁰ and shapely³¹ libraries and the
1467 featurization classes are available on the SimBA GitHub repository.

1468

1469 *Grooming and rearing behavioral classifiers:* We used the same feature set to create classifiers for
1470 rearing and grooming. This feature set included movement and acceleration measures of individual
1471 body-parts as well as paired body-part movement Spearman correlations in sliding time windows

1472 (0.25-2s). We also computed distribution descriptive statistics (mean, variance, sum, z-scores,
1473 skew, median absolute deviations, mean absolute change, root mean square) of the animal hull
1474 geometry and animal hull sub-geometries (animal hull width, length, and area, as well as head area,
1475 posterior hull area, anterior hull area, left lateral hull area, right lateral hull area) in sliding windows
1476 of 0.25-2s. We computed Spearman correlations between the hull sub-geometry areas in the same
1477 sized sliding time-windows. Using circular statistics, we calculated descriptive statistics of animal
1478 directions including animal circular range, animal circular standard deviation, and directional vector
1479 lengths in sliding windows of 0.25-2s.

1480

1481 *Straub tail classifier*: For accurate Straub tail segmentation, we first performed video background
1482 subtraction and centered and egocentrically aligned the mice in the videos. Next, we defined a
1483 buffered (1.75cm in each direction) polygonal geometry using the mouse tail key-points (tail-base,
1484 tail center, tail-tip) and sliced the tail geometry from each frame. Using the tail geometry
1485 coordinates, we computed the tail geometry mean area, standard deviation, mean absolute deviation
1486 in sliding time-windows of 1-2s. For each frame, we compared the tail geometry to the tail
1487 geometry 0.5-2s seconds prior using Hausdorff distances. For the sliced images of the tails, we
1488 compared the current tail image versus the tail image 0.5-2s prior using mean squared error of the
1489 raw RGB pixel values. We also computed descriptive statistics (average, variance, sum, mean
1490 absolute change) of the tail tip and tail center movements, and the tail length in sliding windows of
1491 1-2s.

1492

1493 *Feature importance*: Model Gini feature importance scores showed the following key feature
1494 determinants for each behavior: (1) *Grooming* was primarily influenced by features describing the

1495 distribution of the animal's head area, body length, and snout movement within sliding temporal
1496 windows. (2) *Rearing* was primarily driven by features capturing the distribution of the distance
1497 between the animal's ears, head area, and body width in sliding temporal windows. (3) *Straub tail*
1498 was mainly determined by features describing the distribution of the tail-tip movements, tail length
1499 and area, overall body area and right-side hull area, as well as correlations between the movements
1500 of the tail tip and tail base in sliding temporal windows. See [Table S2](#) for detailed feature
1501 importance.

1502

1503 *Performance:* Out-of-sample frame wise performance showed strong performance for grooming
1504 (precision: 0.92, recall: 0.99, f1: 0.96), rearing, (precision: 0.98, recall: 0.96, f1: 0.97) and Straub
1505 tail (precision: 0.99, recall: 0.96, f1: 0.98). See [Table S2](#) for model metrics.

1506

1507 *Heuristic models:* Freezing and circling were detected using heuristic rules as previously
1508 described^{32,33}. Circling was scored as present when the directional circular range of the animal was
1509 above 320 degrees, and the animal movement (computed from the animal centroid key-point) was
1510 above 6cm in the preceding 10s time-window. Freezing was scored as present in frames where the
1511 velocity (computed from the mean movement of the nape, snout, and tail-base body-parts) fell
1512 below 5 mm/s in the preceding 3s. Although the nape body-part was not pose-estimated, this body-
1513 part was approximated to be halfway between the two ears.

1514

1515 *Wireless mechano-acoustic device implant:* Mice were subdermally implanted with wireless
1516 mechano-acoustic devices (MA device: nVital, NeuroLux, Inc.) on the ventral surface while under
1517 isoflurane anesthesia for a duration of less than 30 min. They also received single subcutaneous

1518 injections of lidocaine-bupivacaine and ketoprofen for peri-operative analgesia. They were then
1519 group housed following surgery. The devices contain an accelerometer adhered to the animal's
1520 sternum with GLUture, mechanically coupled to cardiac, respiratory, and behavioral motion, as
1521 well as a temperature sensor. Mechano-acoustic and temperature signals stream via Bluetooth at a
1522 rate of 800Hz and 1Hz, respectively. Wireless power transfer by resonant magnetic inductive
1523 coupling allows the devices to wirelessly charge and transmit data. See Ouyang et al. (2024)³⁴ for
1524 surgical procedure and device details.

1525

1526 *MA physiological data processing & analysis:* For heart rate, first heart sounds were found by
1527 subtracting the behavioral motion component of the z acceleration signal, as estimated by a
1528 Savitzky-Golay filter (8th order, 21-point window), from the raw z acceleration. The lowpass
1529 filtered (15Hz cutoff) Shannon energy envelope of the heart sound signal revealed individual heart
1530 beats. Average beat-to-beat intervals per second yielded heart beats per minute. The respiratory
1531 signal was found by down-sampling z-axis acceleration to 50Hz, and band-pass filtering (2 - 5 Hz
1532 cutoff) revealed individual respirations. Average beat-to-beat intervals per second yielded
1533 respirations per minute. Physical activity was calculated by finding the average variance of each
1534 acceleration axis (normalized to system noise variance) in 1 second rolling windows and taking the
1535 square root to quantify acceleration variability in arbitrary units. bpm: beats-per-minute; rpm:
1536 respirations-per-minute; a.u.: arbitrary units. See Ouyang et al. (2024)³⁴ for further details.
1537 Statistical comparisons were performed on raw (unsmoothed) physiological metrics averaged across
1538 a predefined 5–15 min post-injection window. Normality of paired differences per subject was
1539 assessed using the Shapiro-Wilk test. Differences between chemogenetic modulation conditions
1540 (CNO vs saline) were tested using one-tailed t-tests. Data processing was performed using custom

1541 Python scripts (scipy.stats), and group means and paired t-tests were conducted in GraphPad Prism
1542 (see [Table S3](#) for detailed statistical outputs).

1543

1544 *Modified open field test and MA device data collection for isoflurane induction* ([Fig 3](#)): A 15cm x
1545 15cm smaller, modified open field box was adapted with an isoflurane in oxygen delivery inlet, gas
1546 analyzer sampling line, and scavenge. The sides of the box were wrapped in 14G copper wire
1547 attached to tuner and power distribution control (PDC) boxes to charge the implanted devices and
1548 allow data streaming (NeuroLux, Inc). 7 animals (2 trials each) were recorded under isoflurane
1549 exposure. Trials consisted of 5 minutes of recording before isoflurane onset, 20 minutes following
1550 loss of reflex to a loud clap, and 20 minutes after isoflurane delivery ended.

1551

1552 *Modified open field test and MA device data collection for chemogenetic studies* ([Fig 4](#) and [Fig 7](#)):
1553 For 3 days before the experiment, mice were habituated to saline intraperitoneal (i.p.) injections and
1554 explored a 15cm x 15cm modified open field box for 10 min. The sides of the box were wrapped in
1555 14G copper wire attached to tuner and power distribution control (PDC) boxes (NeuroLux, Inc).
1556 Implanted wireless mechano-acoustic (MA) devices (NeuroLux, Inc) were also tested during this
1557 time along with power input from the PDC box (7 watts). During each experiment day, the animal
1558 explored the enclosure for 2-min before an i.p. injection of either saline, 1 mg/kg clozapine-n oxide
1559 (CNO, Enzo Life Sciences), or 5mg/kg CNO. Recordings were taken with a web-camera positioned
1560 directly above the animal for 20-min. The average latency to freezing in all CNO trials was used to
1561 time-align “drug onset” in saline trials. Each mouse underwent 3-6 saline trials and 2-3 CNO trials
1562 with no more than one trial per day. For brain-wide TRAP experiments, a total of 18 CNO trials and
1563 20 saline trials was analyzed from n = 7 Gq+ mice (3 females 4 males) in [Fig 4](#), and a total of 17

1564 CNO trials and 17 saline trials was analyzed from n = 6 Gi+ mice (2 females, 4 males) in [Fig S4](#).
 1565 For LPB TRAP experiments, a total of 29 CNO trials and 47 saline trials was analyzed from n = 10
 1566 Gq+ mice (2 females, 8 males) in [Fig 7](#), and a total of 21 CNO trials and 31 saline trials was
 1567 analyzed from n = 7 Gi+ mice (2 females, 5 males) in [Fig S6](#). Behaviors for a sub-selection of
 1568 animals were manually scored to identify commonly observed behaviors, as well as determine
 1569 criteria for behavioral classifiers. Video recordings collected during the experiment were used for
 1570 pose estimation and supervised behavioral analysis (see [Modified open field test behavioral](#)
 1571 [analysis](#)) and analyzed for changes in maximum velocity using ANY-maze video tracking software
 1572 ([Fig S4](#) and [Fig S6](#)). Data collected from the MA devices were processed with custom Python
 1573 scripts. GraphPad Prism was used to analyze group means and paired t-tests, and to analyze data
 1574 with 2-way ANOVA and Tukey multiple comparison post-tests. See [Table S3](#) and [Table S5](#) for
 1575 statistics.

1576

1577 *Perfusion for tissue processing:* After all studies, mice were deeply anesthetized with isoflurane and
 1578 underwent transcardial perfusion with phosphate-buffered saline (PBS) followed by 10% formalin
 1579 and dissection of the brain for further analysis (see [Intact brain clearing and immunolabeling](#)).

1580

1581 *Modified open field test behavioral analysis:* We created pose estimation models to track 9 body-
 1582 part key-points using SLEAP³ (snout, left ear, right ear, left lateral, right lateral, centroid, tail base,
 1583 tail center, and tail-tip). Missing pose-estimation data was interpolated using forward fill and the
 1584 data was smoothed using a Savitsky-Golay filter across a 500ms sliding window. Grooming,
 1585 rearing, and Straub tail were detected using random forest classifiers⁴. Freezing and circling were
 1586 detected using heuristic rules as previously described^{32,33}.

1587

1588 *Electrocorticography surgical procedure & neural signal data recording:* Electrocorticography
 1589 (ECoG) was recorded using a tethered implant as previously described²⁹. Briefly, using aseptic
 1590 stereotaxic surgical techniques under isoflurane anesthesia, a midline incision was made above the
 1591 skull, and ECoG-recording electrodes (Pinnacle Technology, Lawrence, KS; No. 8209: 0.10-in.)
 1592 were screwed through cranial holes at the following coordinates:

- 1593 ❑ *Channel 1 (left frontal cortex):* 1.5 mm lateral and 2 mm anterior to bregma
- 1594 ❑ *Channel 2 (right parietal cortex):* 1.5 mm lateral and 2 mm posterior to bregma
- 1595 ❑ *Ground (visual cortex):* 1.5 mm lateral and 4 mm posterior to bregma
- 1596 ❑ *Reference (cerebellum):* 1.5 mm lateral and 6 mm posterior to bregma

1597 Mice were singly housed in standard cages under a 12:12 light-dark (LD) cycle following
 1598 surgery. After a 14-day recovery period, they were moved to recording cages, fitted with a tether
 1599 and preamplifier (100x signal gain) connected to the Pinnacle Technology recording system, and
 1600 given at least 1 day to acclimate before recording.

1601 Animals were injected with saline or CNO (5mg/kg) at the same time of day on consecutive
 1602 days (days 1 and 2, respectively), with pre-experimental (day 0) and post-experimental (day 3) days
 1603 serving as baseline comparisons. To assess neural dynamics during isoflurane, mice were placed in
 1604 a specialized chamber and induced with 3% isoflurane until fully immobile (~10 minutes).
 1605 Isoflurane was then discontinued, and the mice were allowed to fully wake (~20–30 minutes). In all
 1606 recording days, electromyography (EMG) signals were obtained by inserting a pair of silver wires
 1607 into the neck muscles. All screws and wires were connected to a common 6-pin connector
 1608 compatible with the Pinnacle recording device using silver wire. Signals were continuously
 1609 recorded at 400 Hz with low-pass filters at 100 Hz. All recordings were saved and exported for

1610 processing using the European Data Format (EDF). Simultaneously, animal activity was
1611 continuously recorded using webcams.

1612 For each animal, one hour of data (time-matched across days) was extracted for each
1613 condition (pre, saline, CNO, post), along with the entirety of the isoflurane induction condition, and
1614 saved to individual .edf files. Spectrograms were generated by low-pass filtering each file at 12 Hz
1615 and log-transforming the signal data. Additionally, slow wave signal-to-noise ratio (SNR) was
1616 calculated using minute-long bins by dividing mean power in the delta frequency band (1–4 Hz,
1617 primarily capturing delta waves associated with sleep and anesthesia³⁵) by mean power in
1618 surrounding frequencies (0.5–1 Hz + 4–7 Hz). All analyses were conducted using the left frontal
1619 cortex electrode (i.e., channel 1).

1620 To quantify the change in delta (1-4 Hz) over the course of drug onset (i.e., 1 minute prior to
1621 and 15 minutes following drug onset), we ran a linear mixed model predicting delta activity ratio
1622 using the interaction term *time* × *condition* (using the ‘Pre’ condition as reference and controlling
1623 for the random effect of individual animal). The results show that the only significant differences in
1624 delta activity were the increases seen over time in the CNO and ISO conditions, with no other
1625 significant direct or interaction effects (see [Table S3](#)).

1626
1627 *Head-fixed implantation and habituation:* For high density silicon probe (Neuropixels) recordings,
1628 mice were implanted with a custom-made titanium headplate (H.E. Parmer Company, TN) for
1629 head-fixation of the animal, and 3D-printed “well”³⁶ providing safe access for probe insertion and
1630 retention of saline solution on the skull surface during recordings. Following recovery (~2 weeks),
1631 mice were habituated to head-fixation for at least 3 days, starting with 15- to 20-min periods and
1632 increasing daily by 20 min until they exhibited minimal distress with fixation for 1 h³⁷. For head-

fixed recordings, mice were placed in a 3D-printed mouse holder³⁸ adapted for isoflurane delivery via a 3D-printed stereotax anesthetic nose cone. During recordings, isoflurane gas concentration was measured with a gas analyzer (BC Biomedical, MO).

High-density in vivo electrophysiology: At least 1 day prior to each recording, craniotomies were made targeting regions of interest, planned beforehand using a specialized tool³⁹ to take into consideration optimal probe orientations and angles of entry (5-15°), and finally sealed with Kwik-Cast (World Precision Instruments, FL). When not sealed, craniotomies were kept moist with continuous applications of saline into the “well.” On recording days, mice were first head-fixed and allowed to acclimate before probe insertion. Neuropixels probes were mounted on a micromanipulator system (Sensapex, SC) and lowered via the planned trajectory to 100 μm beyond the target depth, then retracted after a 10-min delay and allowed to rest for an additional 10 min before starting the baseline recording. All recordings were acquired using SpikeGLX (Janelia Research Campus, VA). Recordings (**Fig 6A**) began with a 5-min awake baseline period, followed by isoflurane exposure (ISO, 1.3% target concentration on the vaporizer) maintained for at least 20 min. During induction, mice were kept warm via external heating, and loss-of-reflex (LOR) was verified via intermittent foot pinches. Following the 20-min ISO phase, the vaporizer was turned off to initiate a 25-min washout period, followed by a second 5-min awake baseline recording. Mice were subsequently injected with CNO (5 mg/kg, i.p.) to induce activation of isoflurane-tagged Gq⁺ neural ensembles, and recorded for an additional 30 min post-injection to monitor onset response activity. All recordings were acquired using SpikeGLX (Janelia Research Campus, VA).

1655 *Histological alignment of recording tracts:* To determine the placement of isolated units in the
 1656 brain, sorted units were aligned to the probe tracts after intact brain imaging. After recordings,
 1657 intact formalin-fixed brains were cleared and imaged with light sheet fluorescent microscopy for
 1658 probe tract tracing in the Allen Brain Atlas space and mapped onto the locations of isolated units to
 1659 identify their placement in the brain (**Fig 6B** and see *Intact brain clearing* and *imaging*). After
 1660 registering each brain to the Allen CCF¹¹, probe tracts made by application of the lipophilic CM-DiI
 1661 dye (Thermo Fisher, WA) to the probe shank before each recording were traced in Lasagna¹⁰ to
 1662 yield a fitted line of XYZ coordinates, which were visualized in Brainrender¹¹. A separate open-
 1663 source tool⁴⁰ was used to adjust the alignment of activity recorded along the shank to the regions
 1664 identified along the tract.

1665

1666 *Electrophysiological data analysis:* Spike sorting was performed with Kilosort4⁹, isolated units
 1667 were discarded if they had fewer than 500 spikes across the recording and filtered out further based
 1668 on quality metrics assessment⁴¹. Quality metric thresholds were adjusted from standard defaults to
 1669 account for neurons rendered silent during burst suppression epochs, as well as interrupted spike-
 1670 train continuity leveraged for waveform drift correction, a potential cause for over-splitting^{42,43}.
 1671 Resulting units were then inspected for post-hoc merging during manual curation. Postprocessing
 1672 and curation were performed using custom scripts integrating open-source Python packages⁴⁴.
 1673 Spike times were binned (250 ms) and per-sec firing rates were log10-transformed, then z-scored
 1674 relative to the initial baseline period, with a regularization term added to the standard deviation to
 1675 stabilize low-firing units. For classification of responses, each unit was classified as activated,
 1676 suppressed, or neutral for ISO and CNO based on their change in z-scored activity (delta Z) during
 1677 a post-onset period, relative to pre-onset (ISO: -3 min to ISO onset vs. 3 to 6 min post ISO onset;

1678 CNO: -5 min to CNO injection vs. 15 to 20 min post CNO injection). Within each comparison
1679 period, windows were re-binned into 5 s bins and compared per-unit using Wilcoxon signed-rank
1680 tests. FDR-corrections (Benjamini-Hochberg) were applied across all units, which were considered
1681 “responsive” at $q < 0.05$, and defined as “activated” vs. “suppressed” depending on the sign of their
1682 delta Z. To establish significant response rates above chance, we first obtained empirical nulls via
1683 circular-shift permutations for comparison (500 shuffles), shifting each unit’s combined pre- and
1684 post-period window bins by a random offset (between 10-90% of window), to disrupt the alignment
1685 of activity to experimental onset periods while maintaining intrinsic temporal dynamics. The
1686 classification pipeline was then re-applied to obtain the null distributions of response proportions
1687 for each analyzed brain region. Observed responsiveness rates were then determined to be
1688 significant if they exceeded these region-specific null rates (one-tailed exact binomial tests), and
1689 whether responsive populations exhibited a majority of activated vs. suppressed units (two-tailed
1690 binomial tests; $p=0.5$). For each region, to determine whether ISO- and CNO-responsiveness co-
1691 occurred above chance, we tested overlaps based on Fisher’s exact tests on 2x2 conditional tables
1692 (ISO+ vs. ISO- vs. CNO+ vs. CNO-), across each region’s units. Overlaps are reported as
1693 conditional odds ratio with 95% confidence intervals, and confirmed using permutation tests
1694 shuffling CNO-responsiveness labels across units (10k iterations). Among units responsive to both
1695 ISO and CNO, whether activated (A) or suppressed (S), we applied Fisher’s exact tests on their
1696 direction contingency table (ISO-CNO: A-A vs. A-S vs. S-A vs. S-S) to determine whether
1697 response direction was maintained, with same-direction pairs (A-A, S-S) over-represented
1698 compared to opposite-direction (A-S, S-A). To determine whether responsiveness overlap was
1699 particularly enriched in PB relative to non-PB hindbrain, we fit a logistic model across all hindbrain
1700 units predicting ISO responsiveness from their CNO responsive label and whether they were in PB,

1701 to test whether the overlap in the latter was stronger than in neighboring hindbrain regions. Given
 1702 small unit counts for the region, significance was defined using a penalized likelihood-ratio test. All
 1703 analyses were conducted using custom Python scripts integrating open-source packages.

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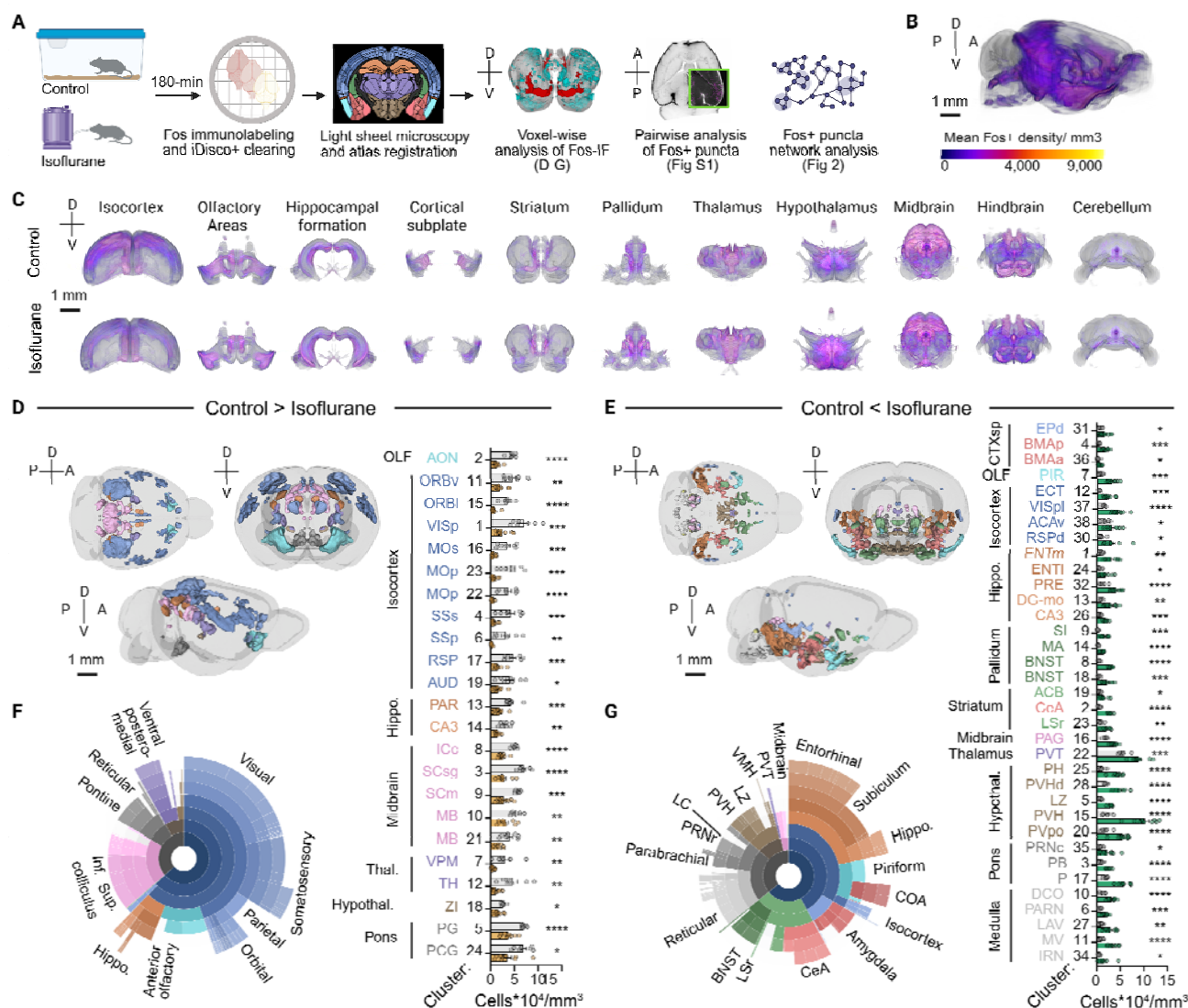


Figure 1: Single-cell-resolution map of isoflurane-activated brain-wide Fos expression.

(A) Experimental overview: Three complementary pipelines were used to analyze whole brain Fos activation in home cage control and isoflurane-treated mice: voxel-wise immunofluorescence analysis with UNRAVEL (panels D-G), atlas-region-based Fos+ puncta quantification with ClearMap (Fig S1), and functional network analysis with SMARTTR (Fig 2). (B) Mean Fos+ density shown as a heatmap across the intact brain in all isoflurane samples (n = 9). (C) Mean Fos+ density heatmap across all control (n = 7, top row) and isoflurane samples (n = 9, bottom row), virtually sliced in the coronal plane at major anatomical subdivisions (see Fig S1 for detailed ClearMap analysis and representative Fos images). UNRAVEL mapped regions where isoflurane either (D) suppressed Fos ($q < 0.05$) or (E) enhanced Fos ($q < 0.005$). FDR-corrected clusters differing in Fos+ cell density are shown in mirrored 3D brains and bar graphs (see Video S1 and Methods). Data are shown as mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, unpaired two-tailed t-tests. (F-G) Sunburst plots summarize the anatomical composition of valid clusters by volume, with outer rings representing finer anatomical granularity in the Allen Mouse Brain Common Coordinate Framework (CCFv3) atlas hierarchy. Abbreviations are defined in Table S1 with these exceptions: {'a': 'anterior', 'd': 'dorsal', 'l': 'lateral', 'v': 'ventral'}. Raw Fos counts and z-scored ClearMap data are provided in Table S1. See Fig S2 and Table S1 for details on each valid cluster and raw cell densities across multiple FDR thresholds. Scale bar, 1 mm.

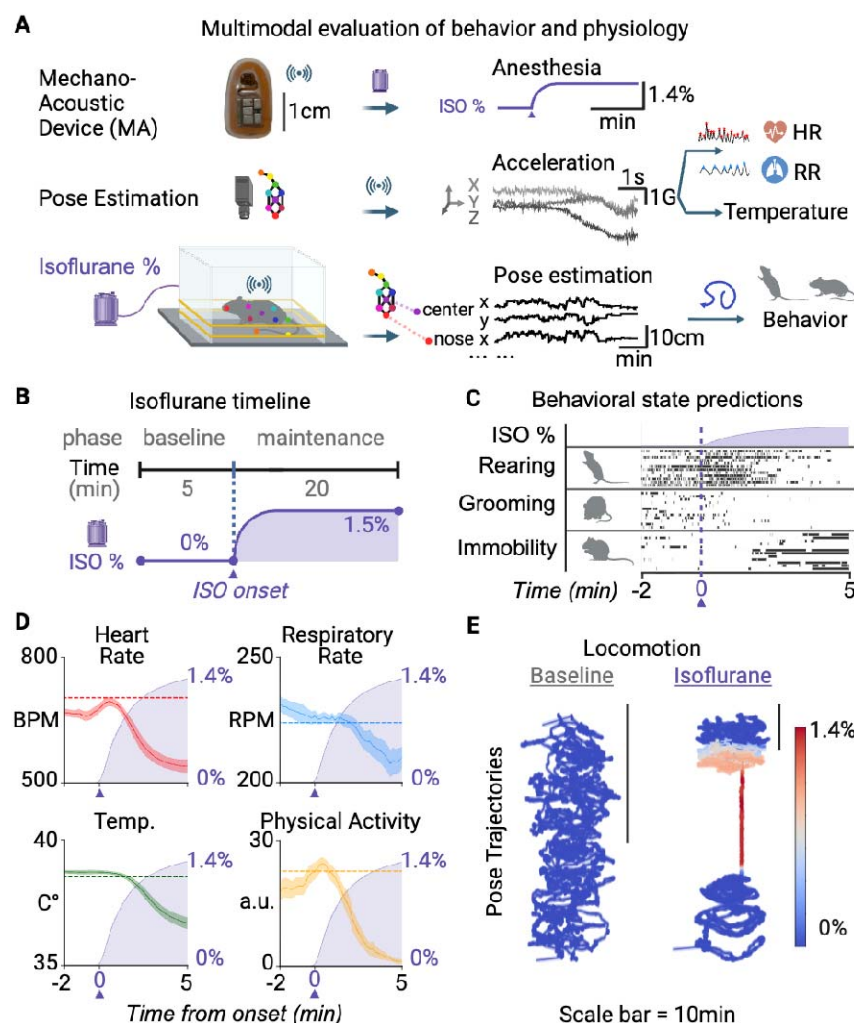


Figure 3. Integration of multimodal behavior and peripheral physiology during isoflurane state transitions.

(A) Mechano-acoustic devices (MA devices) were implanted in mice to obtain internal physiological metrics (heart rate, HR; respiratory rate, RR; temperature & physical activity), integrated with pose estimation and isoflurane (ISO) gas concentration readout for a comprehensive assessment of internal state during isoflurane anesthesia. (B) We obtained physiological measurements as mice ($n = 7$) progressed through isoflurane induction. (C) Behavioral events detected by supervised classifiers using pose estimation. With increasing ISO concentration, mice gradually ceased grooming and rearing behaviors as they exhibited extended immobility. See [Video S3](#) for representative integration of behavior and physiology. (D) Following induction (“ISO onset”), heart rate (“HR”; red), respiratory rate (“RR”; blue), temperature (“Temp.”; green) and physical activity (yellow) all decreased with increasing ISO concentration (%; violet), preceded by a brief rise only in HR and physical activity. Horizontal dashed lines show means of baseline trials (Means $\pm$ SEM; heart rate, 703.0 ± 2.26 bpm; respiratory rate, 223.8 ± 0.56 rpm; body temperature, 38.5 ± 0.01 °C; physical activity, 22.7 ± 0.38 a.u.). See [Table S3](#) for detailed statistics. (E) 3-dimensional plot of pose trajectories reflecting locomotor changes with increasing isoflurane concentration. Color intensity represents ISO concentration from 0 to 1.4%. Scale bar is 10 minutes.

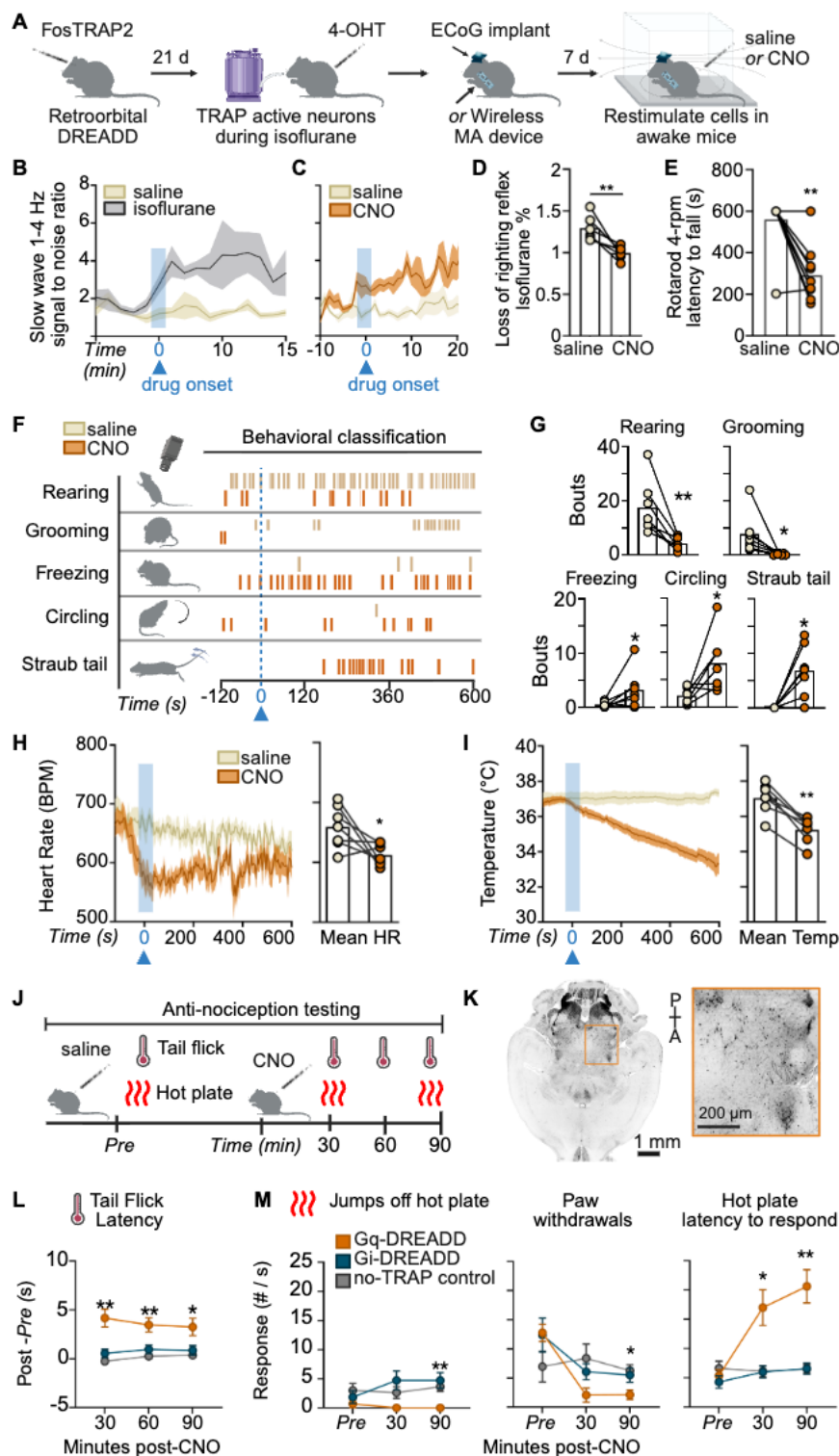


Figure 4: Brain-wide activity-dependent capture of anesthesia unconsciousness.

(A) Experimental timeline: FosTRAP2 transgenic mice received retroorbital injection of AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure (1.2-1.3%) with 4-hydroxytamoxifen (4-OHT); (DREADD = designer receptor engineered to be activated by a designer drug). Mice were tested in the absence of anesthetic after either clozapine-n-oxide (CNO) or saline injection in a modified open field test with concurrent videorecording and peripheral physiology recording using wireless mechano-acoustic (MA) devices, or neural physiology with electrocorticography (ECoG). (B) 1-

4 Hz delta slow wave activity is prominent after isoflurane induction and (C) after CNO injection as compared to saline in Gq-DREADD mice ($n = 4$, see [Fig S4B](#) for spectrograms). (D) Gq-DREADD expressing mice fall off a rotarod (4-rpm) faster after CNO injection ($n = 10$). (E) Gq-DREADD mice display loss of righting reflex at a significantly reduced isoflurane concentration after CNO injection, though this does not occur at baseline (see [Fig S4C](#) for dose-response and non-TRAP oil control). (F) A representative raster plot from all 6 (3 saline, 3 CNO) of one animal's trials after analysis of classified behavior (also see [Video S4](#)). (G) Bouts of classified behavior for the average of all saline and CNO trials per animal ($n = 7$ mice, total of 18 CNO and 20 saline trials). (H) Left: Heart rate (beats per minute, BPM) decreases after CNO injection as compared to saline, shown time-aligned to drug onset, right: mean heart rate (HR) decreases after CNO injection in Gq-DREADD mice. (I) Left: Body temperature (Celsius) changes over duration of trial, time-aligned to drug onset, right: mean temperature decreases with CNO. See [Fig S4](#) and [Video S4](#) for Gi-DREADD data. (J) Experimental timeline for anti-nociception testing: FosTRAP2 mice received retroorbital injection of either Cre-dependent activating Gq-DREADD or inhibitory Gi-DREADD. They underwent isoflurane exposure, during which either 4-OHT or oil vehicle (if non-TRAP control) was administered. On test day, mice were tested for anti-nociception following saline injection (Pre) and 5 mg/kg CNO (Post). Warm water tail withdrawal (tail flick) latency was tested at 30-, 60-, and 90-minutes (min) post-CNO injection, and hotplate behavior was tested at 30- and 90-min post-CNO injection interleaved with tail flick tests. (K) Representative hindbrain light sheet image after intact brain iDisco+ clearing and mCherry-immunolabeling for DREADD expression ($n = 7$ mice). Scale bar is 1 mm and 200 μ m in inset. See [Fig 5](#) for brain analysis. (L) Gq-DREADD mice ($n = 8$, orange) show increased latency to tail flick compared to Gi-DREADD ($n = 7$, blue) or non-TRAP control mice ($n = 8$, gray) at every time point tested. (M) In the hot plate test, Gq-DREADD mice do not jump off the hotplate, have fewer paw withdrawals off the hot plate at 90-min, and show increased latency to first paw withdrawal response at 30 and 90-min post-CNO. 2-way repeated measures ANOVA with Tukey's multiple comparisons: $*p < 0.05$, $**p < 0.01$. Paired t-tests: $*p < 0.05$, $**p < 0.01$. Error bars are the mean $\pm$ SEM. See [Table S3](#) for detailed statistics.

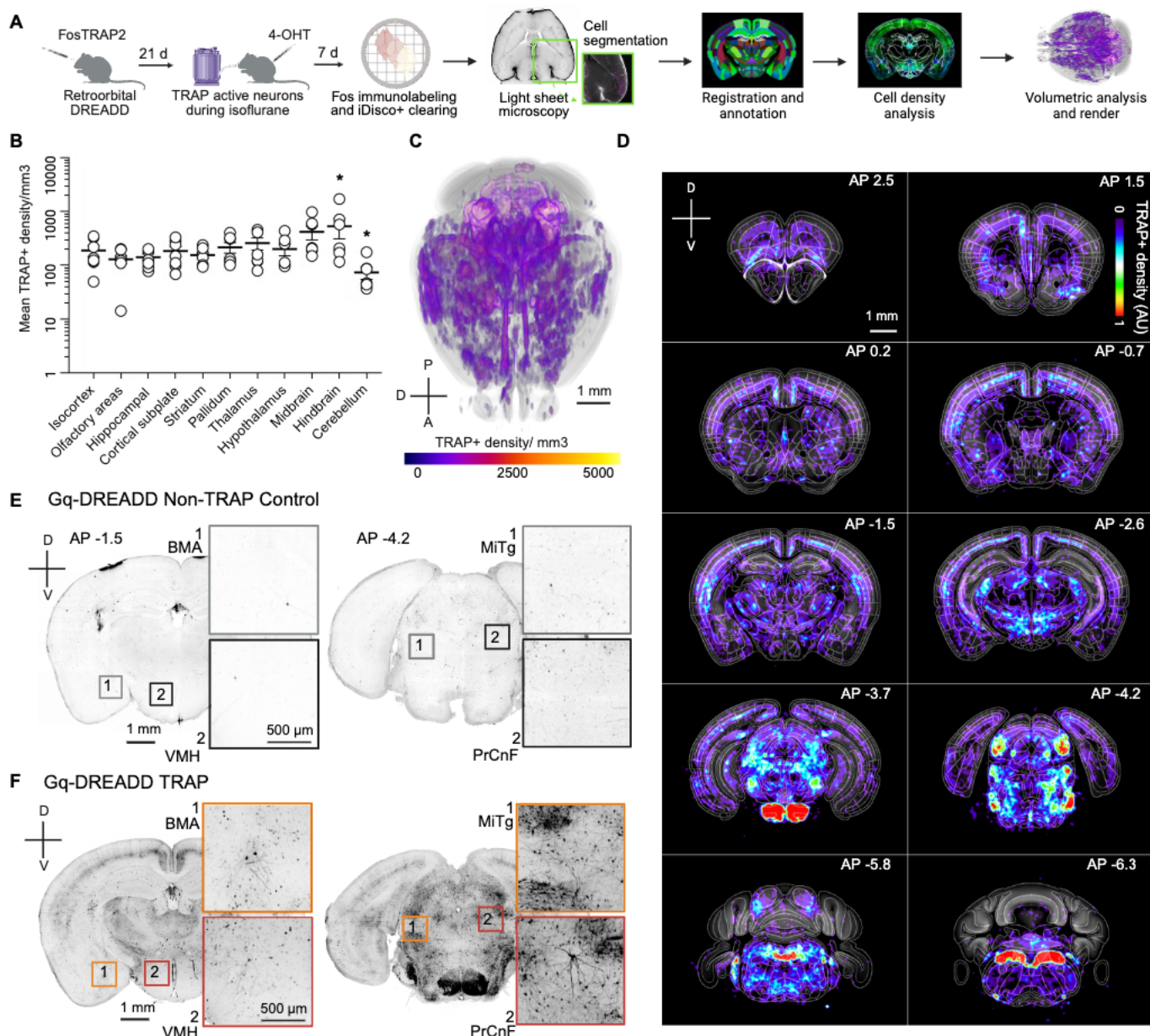


Figure 5. Spatial distribution of DREADD immunolabeling after brain-wide activity-dependent capture. (A) Experimental timeline: FosTRAP2 transgenic mice received retroorbital injection of an AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure with 4-hydroxytamoxifen (4-OHT). After experiments, intact brains were immunolabeled for mCherry-DREADD and cleared by iDisco+ (n = 7 mice). ClearMap was used to register brains, detect TRAP+ signal, generate cell counts and density heat maps. (B) Mean TRAP+ signal density calculated across larger brain subdivisions after atlas registration by ClearMap shows similar TRAP+ signal density across the brain, except for significant differences in the hindbrain and cerebellum (1-way ANOVA, *p<0.05). (C) 3-dimensional mean density heatmap of all samples (n=7). (D) Mean TRAP+ density across all samples shown as voxel-based 2D-projection heatmaps (calibration bar is arbitrary units). (E) Representative whole brain light sheet microscopy images after iDisco clearing and mCherry-DREADD immunolabeling digitally sliced to the 2D coronal plane from one control mouse expressing retroorbital AAV-PHP.eB Cre-dependent Gq-DREADD with oil vehicle administered during the isoflurane exposure (non-TRAP control, see [Methods](#)) and (F) one Gq-DREADD-expressing mouse that received 4-OHT during isoflurane exposure for activity-dependent capture of synthetic unconscious state. Abbreviations: BMA = basomedial amygdala, VMH = ventromedial hypothalamus, MiTG = microcellular

tegmental nucleus, PrCnF = precuneiform area. Scale bar for full coronal slices is 1mm, inset scale bar is 500 μ m. See [Table S3](#) for detailed counts and statistics.

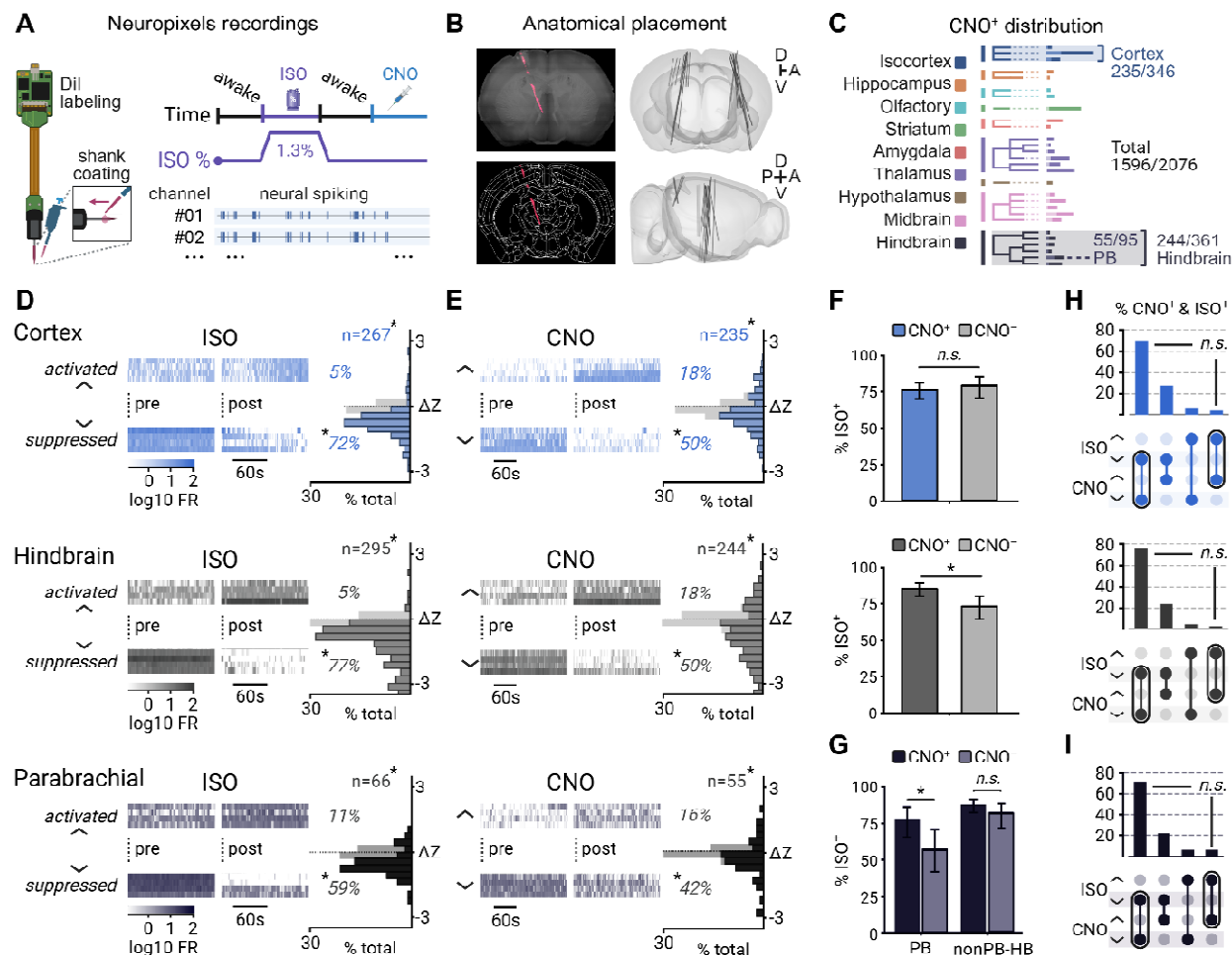


Figure 6: Brain-wide chemogenetic activation preferentially recruits isoflurane-activated neurons in the parabrachial nucleus.

(A) Experimental schematic. Neuropixels probes (DiI-coated for histological tract reconstruction) were used to record single-unit activity from FosTRAP2 mice expressing Gq-DREADD, across a 5-min awake baseline and isoflurane induction (ISO, 1.3%) maintained for at least 20 min. This was followed by a 25-min wash and second 5-min awake baseline before delivering CNO (5 mg/kg) and monitoring chemogenetic activation of ISO-activated ensembles for 30 min. (B) Anatomical localization of recording probe tracts. Left: representative maximum intensity projection (top) across coronal depth (500μm) showing DiI probe tract (magenta), which are traced and aligned to anatomical regions in atlas space (bottom), to determine probe trajectories in the brain (right). (C) Anatomical distribution of recorded units by major region (color key) and the proportion responsive to CNO (darker bar color) per region; isocortex (Cortex, 235/346 CNO-responsive), hindbrain (244/361), and parabrachial nucleus (PB, 55/95) were targeted for analysis. Total CNO-responsive N = 1,596/2,076 total units obtained across 14 recordings, from 8 mice. (D) Left: representative firing-rate rasters (log₁₀ FR) for cortex (top), hindbrain (middle) and parabrachial nucleus (bottom) units significantly activated (top, “□”) or suppressed (bottom, “□”) by isoflurane, showing activity during “pre” (3 min leading up to

onset) and “post” (3 to 6 min after onset) time periods. Right: distribution of single-unit response magnitudes as a change in their baseline-normalized $\log_{10}$ firing rate (ΔZ , post – pre), as a % of total region population; gray bars denote neutral response distribution. **(E)** CNO-response panel as in **D** for cortex (top), hindbrain (middle) and parabrachial nucleus (bottom). Representative raster plots compare activity 5 min leading up to (“pre”) vs. 15-20 min after (“post”) CNO injection. ISO drove significant responsiveness above empirical null in all regions (CTX 267/346, 77.2%; HB 295/361, 81.7%; PB 66/95, 69.5%; $q < 0.0001$), with responsive units strongly biased toward suppression (of ISO-responders: CTX 248/267, 92.9%; HB 278/295, 94.2%; PB 56/66, 84.8%; $q < 0.0001$). Likewise, CNO engaged a significant proportion in all regions (CTX 235/346, 67.9%; HB 244/361, 67.6%; PB 55/95, 57.9%; all $q < 0.0001$), with a majority being similarly biased toward suppression among CNO-responders (CTX 173/235, 73.6%; HB 180/244, 73.8%; PB 40/55, 72.7%; all $q < 0.0001$). **(F-G)** Percentage of units that were ISO-responsive, computed separately within CNO-responsive (CNO+) and CNO-nonresponsive (CNO-) subgroups (mean $\pm$ Wilson 95% CI). **(F)** In cortex (top), ISO+ rates did not differ between CNO+ and CNO- units (179/235, 76.2% vs. 88/111, 79.3%; OR = 0.84, $q = 0.82$), whereas in hindbrain (middle), CNO+ units overall were significantly enriched in ISO+ (209/244, 85.7% vs. 86/117, 73.5%; OR = 2.15, $q = 0.007$). **(G)** The enrichment within hindbrain was significant in LPB (43/55, 78.2% vs. 23/40, 57.5%; OR = 2.65, $q < 0.05$) but not in surrounding non-PB hindbrain (“non-PB-HB”). However, a direct PB vs. non-PB hindbrain comparison (CNO-responsiveness $\times$ region) was in the predicted direction but not significant (Firth-penalized likelihood-ratio test, OR = 1.60, $p = 0.18$). **(H-I)** Upset plots of joint ISO $\times$ CNO classification among double-responsive units, showing the percentage of each direction combination (activated / suppressed). Directional agreement (same sign response across ISO and CNO) did not exceed chance in any region (cortex 70.4%, $q = 0.42$; hindbrain 75.1%, $q = 0.64$; PB 74.4%, $q = 0.32$; nonPB-HB 75.3%, $q = 1$), with same-direction pairs being composed mainly of units suppressed by both. All p-values corrected using Benjamini–Hochberg FDR across 24 tests (6 per region), with significance reported as q values. See [Supplemental Fig S5](#) and [Table S4](#) for detailed statistics and data from additional regions. FR: firing rate; ΔZ : change in baseline-normalized firing rate; OR: odds ratio.

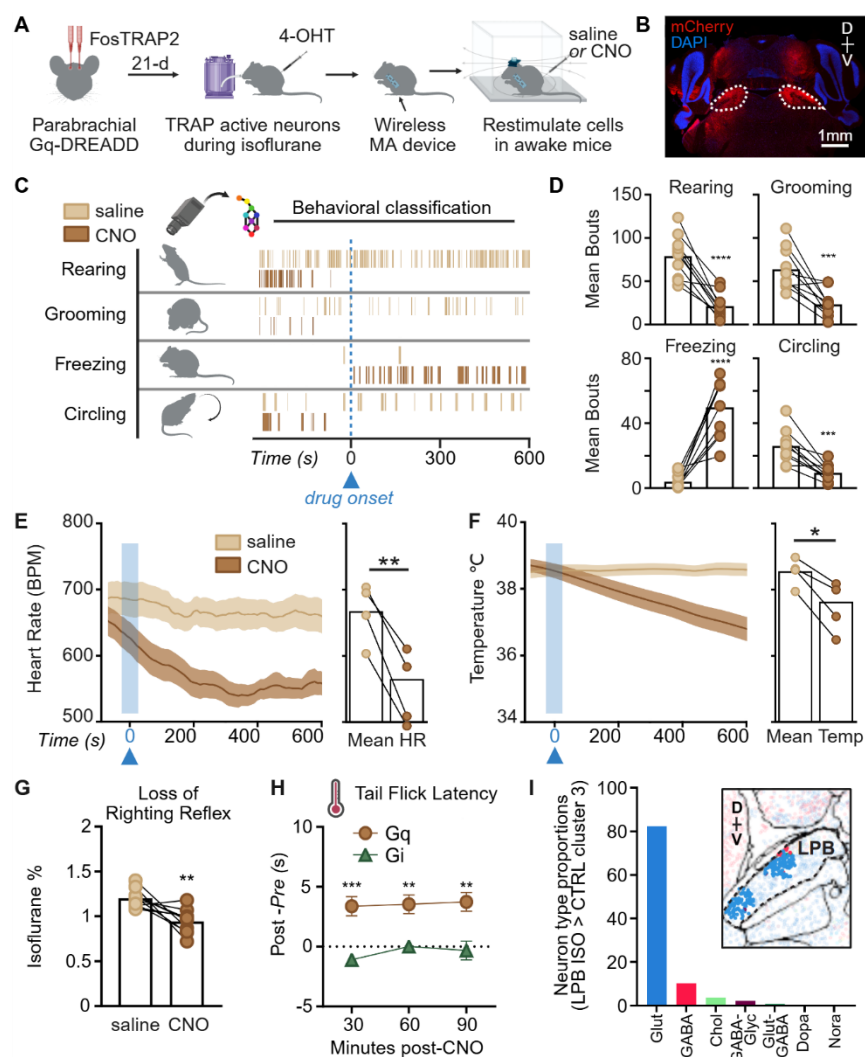


Figure 7: Chemogenetic activity-dependent capture of lateral parabrachial nucleus hub recapitulates components of altered consciousness.

(A) Experimental timeline: FosTRAP2 transgenic mice received bilateral intra-parabrachial injection of AAV-PHP.eB Cre-dependent Gq-DREADD followed by activity-dependent capture (TRAP) during isoflurane exposure (1.2-1.3%) with 4-hydroxytamoxifen (4-OHT); (DREADD = designer receptor engineered to be activated by a designer drug). Mice were tested in the absence of anesthetic after either clozapine-n-oxide (CNO) or saline injection in a modified open field test with concurrent video-recording and peripheral physiology recording using wireless mechano-acoustic (MA) devices. (B) Representative image of mCherry-DREADD immunolabeling after viral expression in the LPB. Scale bar is 1 mm; also see Fig S6. (C) A representative raster plot from one animal's classified behavior from the average of CNO and saline trials (also see Video S5). (D) Bouts of classified behavior for the average of all saline and CNO trials per animal (n = 10 mice, total of 29 CNO and 47 saline trials). (E) Left: Heart rate (beats per minute, BPM) decreases after CNO injection as compared to saline, shown time-aligned to approximated drug

onset, right: mean heart rate (HR) decreases after CNO injection in Gq-DREADD mice ($n = 4$). (F) Left: Body temperature (Celsius) changes over duration of trial, time-aligned to drug onset, right: mean temperature decreases with CNO ($n = 4$). (G) Gq-DREADD mice display loss of righting reflex at a reduced isoflurane concentration after CNO injection ($n = 10$), though this does not occur at baseline. (H) Gq-DREADD mice ($n = 10$, brown) show increased latency to tail flick compared to baseline at every time point tested, while Gi-DREADD mice ($n = 7$, green) do not. (I) LPB cell-types activated by isoflurane analyzed using Fos maps overlaid with Allen Brain Cell Atlas predominantly consist of glutamatergic cells, as well as GABAergic and cholinergic subpopulations; see [Fig S7](#) for detailed analysis. 2-way repeated measures ANOVA with Sidak's multiple comparisons: $**p < 0.01$, $***p < 0.001$. Paired t-tests: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. Error bars represent mean $\pm$ SEM. See [Table S5](#) for detailed statistics. See [Fig S6](#) for all Gi-DREADD data.

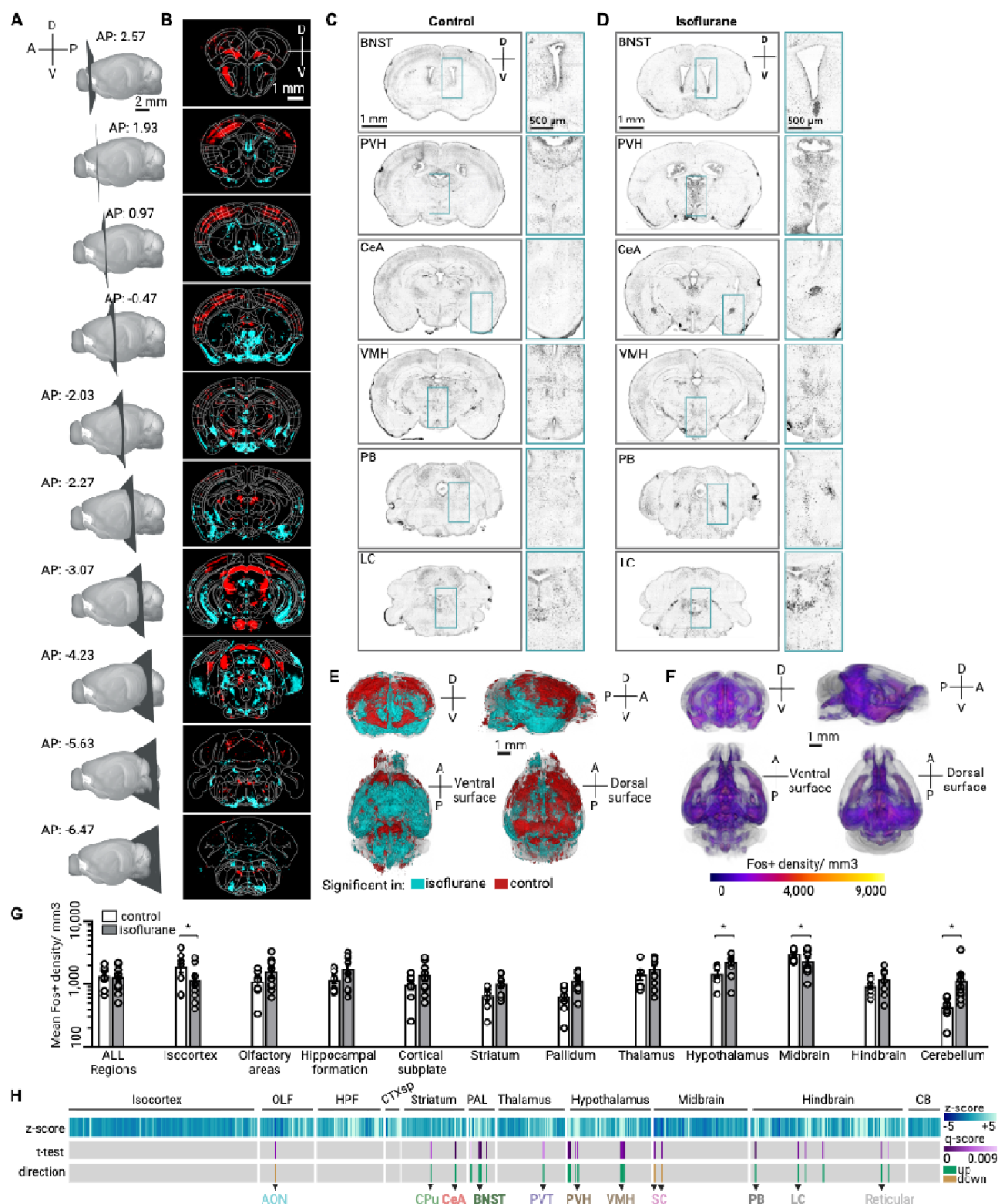


Figure S1. ClearMap analysis and Raw Fos images show subcortical isoflurane-activated hotspots. (A) Intact whole brains were imaged horizontally with light sheet microscopy to acquire cellular resolution Fos+ signal, then digitally resliced in the coronal plane at the indicated AP positions. (B) Voxel-wise comparison of control vs isoflurane conditions, with cyan indicating higher Fos+ cell densities in the isoflurane condition and red indicating higher Fos+ cell densities in the control condition. (C) Representative mean intensity projections of the raw Fos+ signal (3.6x objective) from the control condition and (D) isoflurane condition virtually resliced to the coronal plane at the indicated significant regions, from top to bottom: BNST = bed nucleus of stria

terminalis; PVH = paraventricular hypothalamus; CeA = central amygdala; VMH = ventromedial hypothalamus; PB = parabrachial nucleus; LC = locus coeruleus. **(E)** Voxel-wise significance map of areas significantly regulated in the isoflurane (blue) or control (red) conditions. **(F)** 3D mean Fos+ density heatmaps from the isoflurane condition (also shown in [Fig 1B-C](#)). **(G)** Overall mean Fos+ density, calculated across major brain subdivisions after atlas registration by ClearMap in control vs. isoflurane conditions, identifies significant differences in the isocortex, hypothalamus, midbrain and cerebellum (unpaired t-test, $*p < 0.05$). **(H)** Pairwise analysis of z-scored Fos density by each distinct anatomical subregion after FDR correction ($q < 0.01$) reveals the most significant subregions. Direction of effect in isoflurane relative to control is shown in the bottom bar with green indicating regions of enhanced Fos in isoflurane. Select regions are highlighted (AON: anterior olfactory nucleus, CPu: caudate-putamen, CeA: central amygdala; BNST: bed nucleus of stria terminalis; PVT: paraventricular thalamus; PVH: paraventricular hypothalamus; VMH: ventromedial hypothalamus; SC: superior colliculus; PB: parabrachial nucleus; LC: locus coeruleus). See [Table S1](#) for a full list of regions from pairwise analysis, all raw and z-scored Fos data.

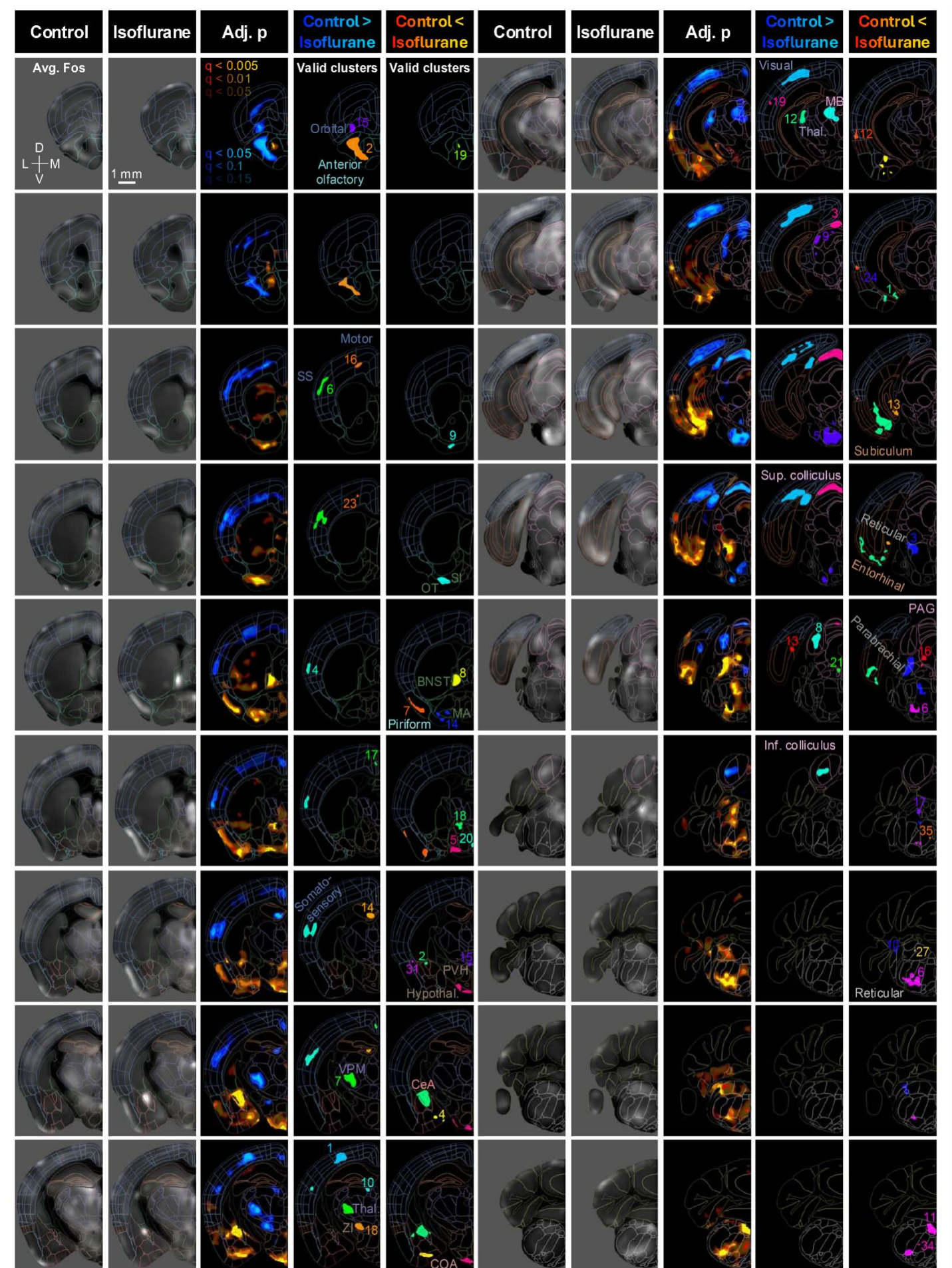


Figure S2. Identification of isoflurane-activated clusters from UNRAVEL analysis.

Voxel-wise data from **Fig 1D-G** were warped to the Allen Mouse Brain Common Coordinate Framework (CCFv3) for display in coronal brain slices and overlaid with a colored wireframe using FSLeyes. Input images for voxel-wise comparisons were averaged for each group to show background-subtracted and z-scored Fos immunofluorescence. FDR-corrected p-value maps show increases (warm colors) and decreases (cool colors) in Fos expression resulting from isoflurane treatment. For each effect direction, the most stringent q-value threshold is fully opaque (100%), the next most stringent is semi-opaque (66%), and the least stringent is less opaque (33%). Valid clusters with significant differences in Fos⁺ cell density were randomly colored and labeled with their IDs. Abbreviations: Adj. = adjusted; Avg. = average; SS = somatosensory; OT = olfactory tubercle; SI = substantia innominata; BNST = bed nucleus of the stria terminalis; MA = magnocellular nucleus; PVH = paraventricular hypothalamic nucleus; Hypothal. = hypothalamus; CeA = central amygdala; Thal. = thalamus; ZI = zona incerta; COA = cortical amygdalar area; MB = midbrain; Sup. = superior; PAG = periaqueductal gray; Inf. = inferior. Please see **Table S1** for a full list of abbreviations and detailed cluster statistics.

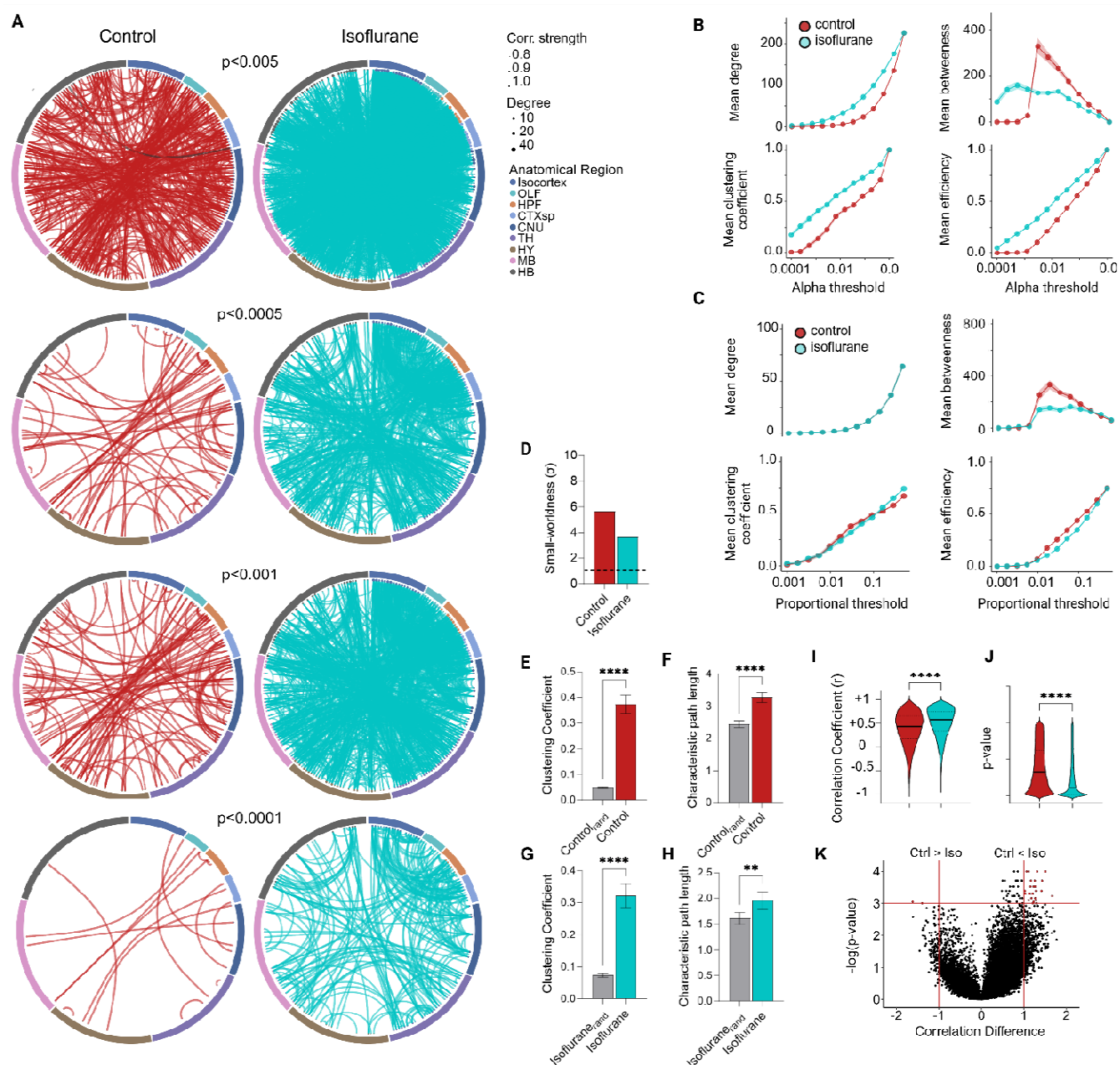
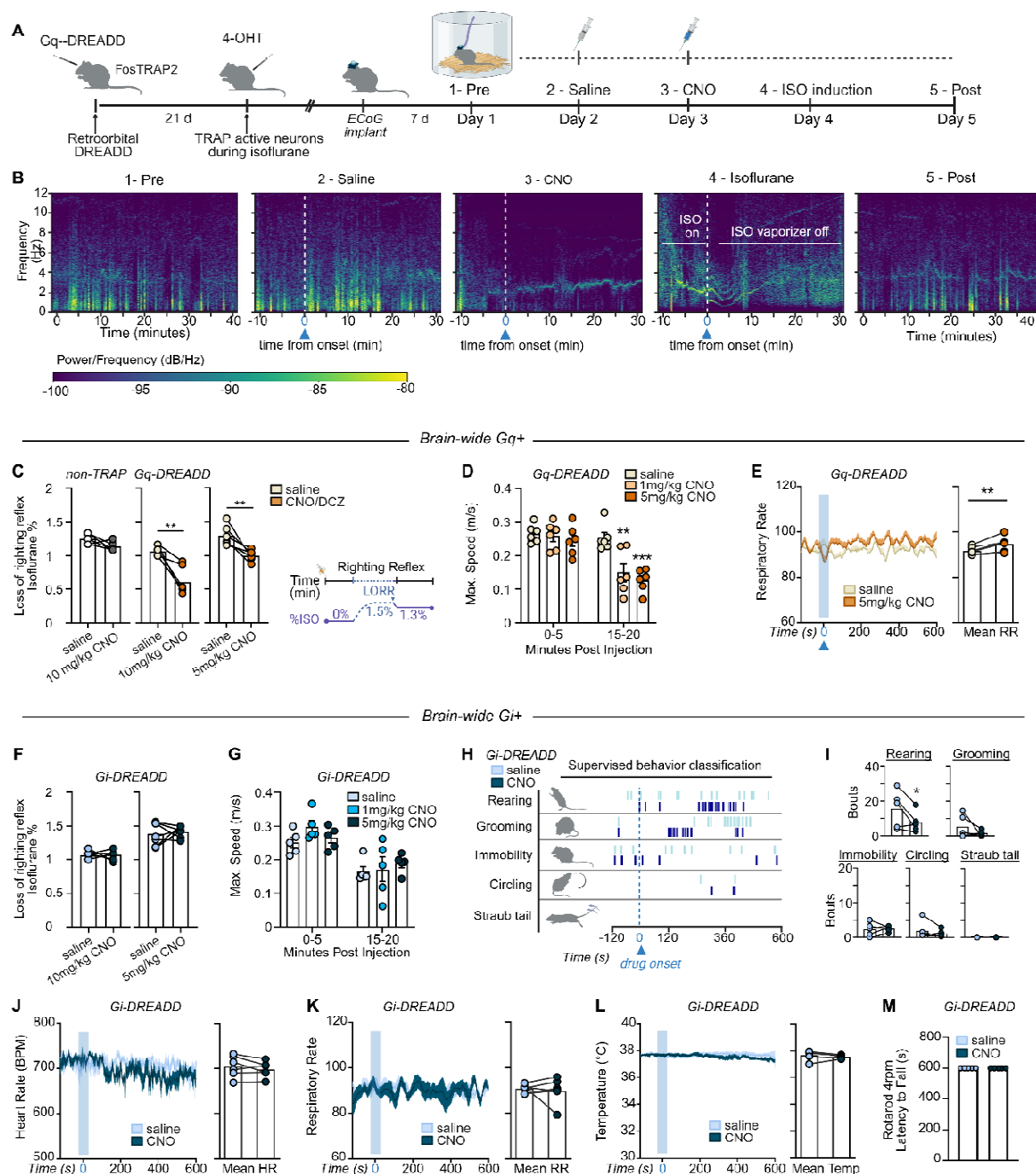


Figure S3. Isoflurane networks are denser across a variety of weight thresholds.

(A) control (red) and isoflurane (blue) networks thresholded at various alpha values. (B) Mean degree, clustering, betweenness, and efficiency metrics calculated across a range of significance thresholds and (C) edge proportion thresholds. (D) Assessment of small-world properties of individual networks plotted in Fig 2. Small-world index (σ) for both networks are above the threshold of 1 (black dotted horizontal line). The control network (E) clustering coefficient and (F) path length as well as the isoflurane (G) clustering coefficient and (H) path length are higher than randomly rewired networks with matching degree sequences. (I) Distributions of the Pearson correlation coefficients between all unique regional connections and their associated (J) p-value for the control and isoflurane networks. (K) Volcano plot illustrating the regional correlations which most differ between the networks. The horizontal red line denotes a significance threshold of $p < 0.001$. Red points in the upper right (control < isoflurane) and upper left quadrants (control > isoflurane), defined by vertical red line lines, indicate regional correlation differences with a magnitude shift greater than 1 between groups. Two-tailed

Student's *t*-test: ** $p < 0.01$, **** $p < 0.0001$. Error bars represent mean $\pm$ SEM. See [Table S2](#) for full abbreviations and detailed statistics.



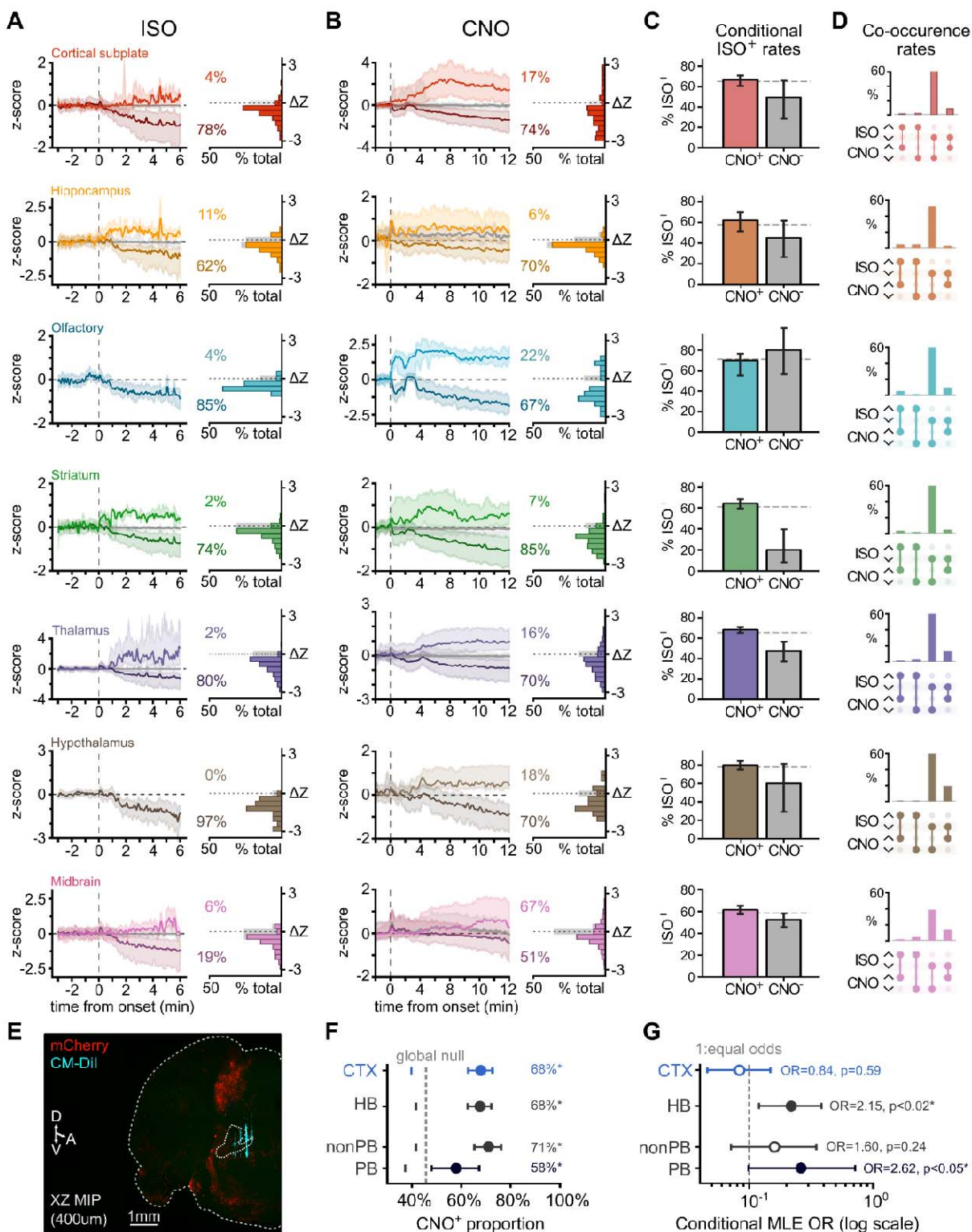
Supplemental Figure S4: Electrocortigraphy and brain-wide chemogenetic behaviors.

(A)

Experimental timeline for electrocortical recordings: FosTRAP2 transgenic mice received retroorbital AAV injection expressing Gq-DREADD and activity-dependent capture during isoflurane

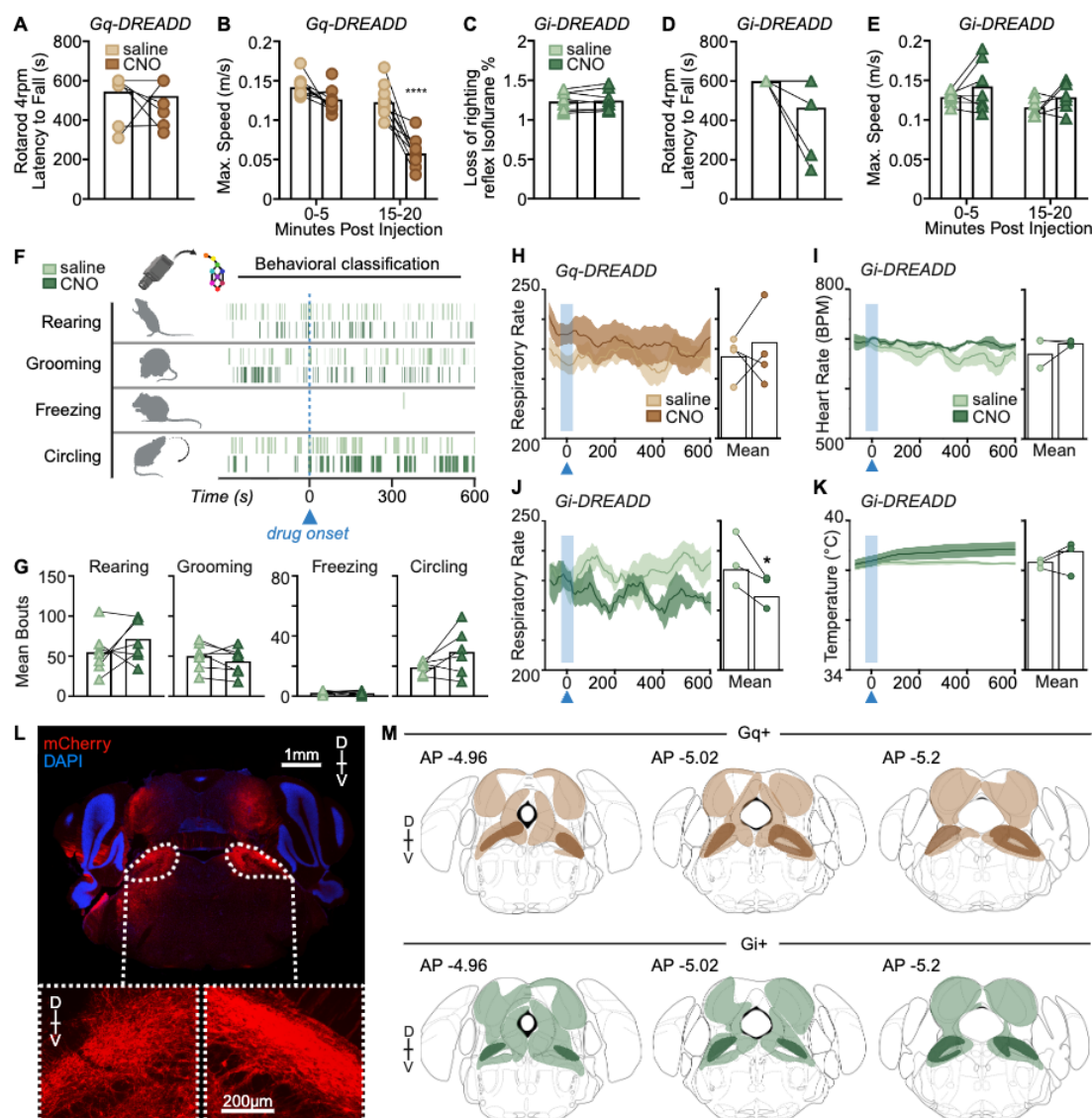
exposure, followed by neural electrocortigraphy (ECoG) recordings over five days (DREADD = designer receptor engineered to be activated by a designer drug, 4-OHT = 4-hydroxytamoxifen, ECoG = electrocortigraphy, CNO = clozapine n-oxide, ISO = isoflurane). **(B)** Day 1 (labeled “1 - Pre”) shows the average 40-minute baseline recording without manipulations. Recordings on Days 2-4 were time-aligned to the approximated onset of intraperitoneal CNO (10 min post-injection) indicated by the white dotted line. Day 5 (labeled “5 - Post”) was obtained without drug manipulation. The spectrograms shown are an average of 4 mice. **(C)** The concentration of isoflurane (%), as measured with a gas analyzer sampling from the chamber, at which mice display loss of righting reflex during saline or CNO (10 mg/kg and 5 mg/kg) is significantly reduced in Gq-DREADD but not non-TRAP controls that received oil injection instead of 4-OHT (n = 5-6 mice). **(D)** CNO

induces dose-dependent reduction in maximum speed (m/s) in the open field in Gq-DREADD mice. **(E)** CNO increases mean respiratory rate in Gq-DREADD mice. **(F)** The concentration of isoflurane (%) at which Gi-DREADD mice display loss of righting reflex during saline or CNO (10 mg/kg, 5mg/kg) is unchanged. **(G)** Maximum speed (m/s) is unchanged in Gi-DREADD mice. **(H)** Representative raster plot from all 6 (3 saline, 3 CNO) of one Gi-DREADD animal's trials after analysis of classified behavior. **(I)** Bouts of classified behavior for the average of saline and CNO (5 mg/kg) trials per animal expressing inhibiting Gi-DREADD virus (n = 5 mice). There is no change in **(J)** heart rate (BPM, beats per minute) **(K)** respiratory rate (breaths per minute) and **(L)** temperature (Celsius) as recorded from the wireless mechano-acoustic device during trials shown in H-I (n = 6 mice, 2-3 trials per drug condition). **(M)** Rotarod (4-rpm) performance in Gi-DREADD mice after 5 mg/kg CNO (n = 5 mice) is unchanged. Error bars show mean $\pm$ SEM. 2-way repeated measures ANOVA with Tukey's post-hoc tests or paired t-tests, *p<0.05, **p<0.01, ***p<0.001. See [Figure 4](#) and [Methods](#) for experimental details. See [Table S3](#) for detailed statistics.



Supplemental Figure S5: Brain-wide TRAP Neuropixels Supplemental Data.

(A) ISO-response panels and (B) CNO-response panels for regions not included for analysis in **Fig 6**. Left: mean baseline-normalized $\log_{10}$ firing rate traces (“z-score”) for units significantly activated (top, lighter) or suppressed (bottom, darker) by isoflurane, shown pre vs. post onset. Right: distribution of single-unit response magnitudes as a change in their baseline-normalized firing rate (ΔZ , post – pre), as a % of total region population; gray bars denote neutral response distribution. (C) Percentage of units that were ISO-responsive, computed separately within CNO-responsive (CNO+) and CNO-nonresponsive (CNO-) subgroups (mean $\pm$ Wilson 95% CI). (D) Upset plots of joint ISO $\times$ CNO classification among double-responsive units, showing the percentage of each direction combination (activated / suppressed). (E) Representative maximum intensity projection (MIP) across coronal depth (400 μ m) showing mCherry-DREADD expression (red) and DiI probe tract (cyan) in parabrachial nucleus (inner dashed outline). (F-G) Forest plots illustrating CI and detailed statistics for (F) proportion tests against region and global empirical shuffle nulls and (G) conditional proportion tests for odds ratio of ISO+ given CNO+ / CNO- (log scale; dashed line at OR = 1, equal odds). See also **Table S4** for detailed counts and statistics. FR: firing rate; OR: odds ratio.



Supplemental Figure S6: Effects of chemogenetic inhibition after activity-dependent capture of lateral parabrachial nucleus hub.

(A) Latency to fall (s) in Gq-DREADD expressing mice is unchanged following CNO injection (5 mg/kg, $n = 10$). (B) Maximum speed (m/s) is significantly reduced in Gq-DREADD mice 15-20 min post-CNO (5 mg/kg, $n = 10$). There is no change in (C) loss of righting reflex (% isoflurane), (D) latency to fall (s) and (E) maximum speed (m/s) in Gi-DREADD mice following CNO injection (5 mg/kg, $n = 7$). (F) A representative raster plot from one animal after analysis of classified behavior (also see [Video S5](#)) after CNO injection as compared to saline in Gi-DREADD mice. (G) Bouts of classified behavior for the average of saline and CNO trials per animal ($n = 7$, total of 21 CNO and 31 saline trials) expressing inhibiting Gi-DREADD virus. CNO had no effect on rearing, grooming, freezing, or circling in Gi-DREADD mice. (H) Respiratory rate in Gq-DREADD mice is unchanged after CNO injection as compared to saline ($n = 4$). (I) Heart rate (BPM, beats per minute) is unchanged following CNO in Gi-DREADD mice ($n = 3$). (J) Respiratory rate is decreased following

CNO in Gi-DREADD mice ($n = 3$). **(K)** Temperature (Celsius) is unchanged following CNO in Gi-DREADD mice ($n = 3$). **(L)** Representative image after mCherry-immunolabeling for DREADD expression and DAPI staining. Scale bar is 1 mm and 200 μm in inset. **(M)** Representative brain sections showing DREADD expression in Gq- (top) and Gi- (bottom) DREADD expressing mice. Highlighted regions indicate areas of viral expression. See [Figure 7](#) and [Methods](#) for experimental details. Error bars show mean $\pm$ SEM. Mixed effects analysis with Sidak's multiple comparisons (panel b, e) and paired t-tests, * $p < 0.05$, **** $p < 0.0001$. See [Table S5](#) for statistics.

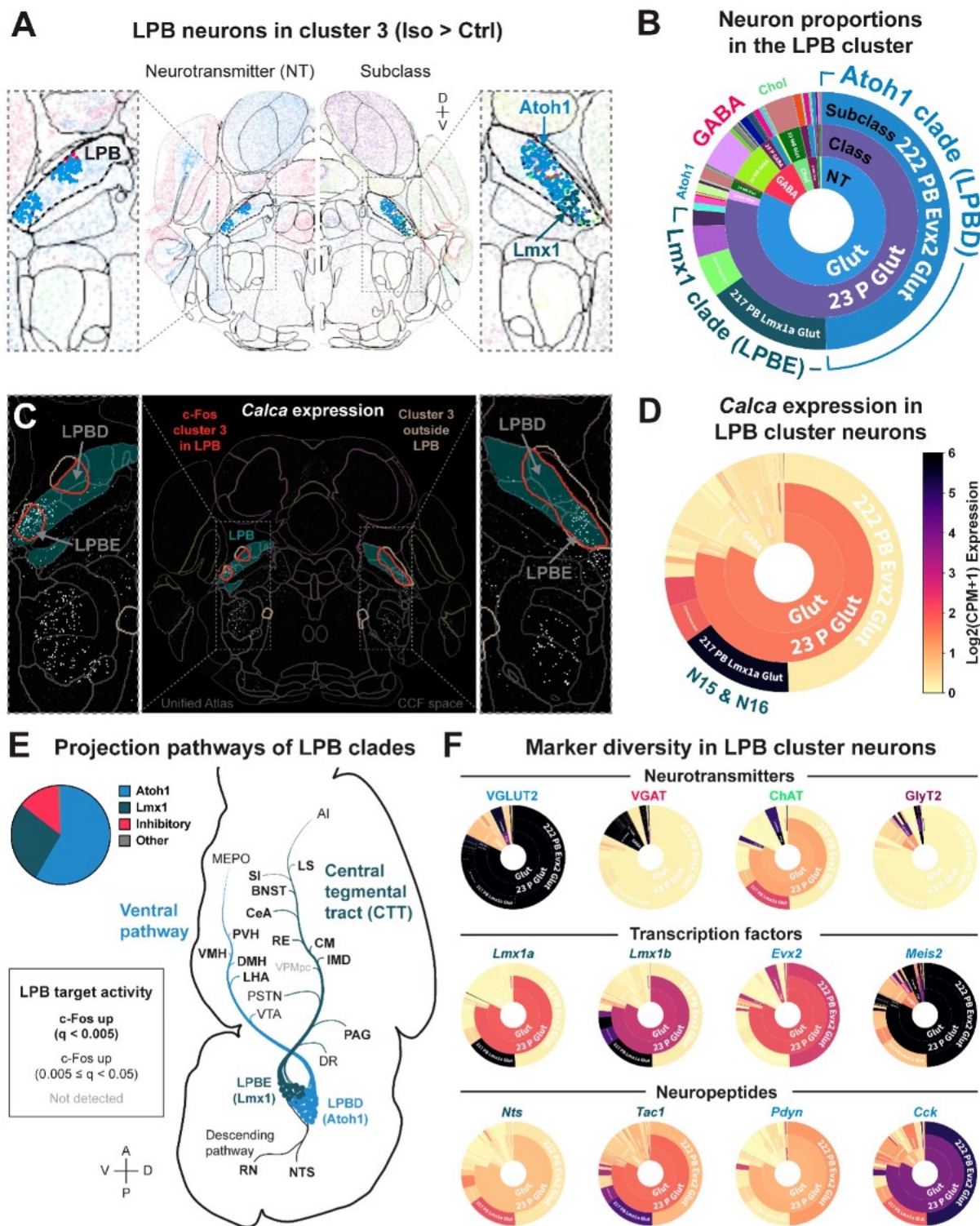


Figure S7. Diversity of neurons in LPB cluster 3.

(A) Neuron types from the Allen Brain Cell Atlas (ABCA) within the LPB portion of Control < Isoflurane cluster 3 (from **Fig. 1D-G**) are highlighted in MERFISH-CCF space. For this, c-Fos

clusters were warped to ABCA MERFISH-CCF space. The spatial distributions of *Atoh1*- and *Lmx1*-derived LPB clades, described by Pauli et al. (2022)⁵¹, are noted for reference. **(B)** ABCA neuron type proportions in the LPB portion of cluster 3. **(C)** To examine the alignment of cluster 3 with LPB subregions, the Unified Mouse Brain Atlas v2 labels were warped to MERFISH-CCF space. LPBE, external LPB; LPBD, dorsal LPB. Imputed Calca expression in ABCA neurons is shown in the context of LPB subregions and cluster 3; display range, 0–6 log₂ (CPM + 1). **(D)** Calca marks CGRP-associated LPB neurons, including N15 and N16 subclusters described by Pauli et al. (2022). **(E)** Major LPB projection pathways adapted from Pauli et al. (2022). *Atoh1*-related LPBD populations project predominantly through the ventral pathway, whereas *Lmx1*-related LPBE populations project predominantly through the central tegmental tract (CTT). Both have additional descending projections. Clade proportions were estimated using imputed marker expression, with marker-positive neurons defined as >3 log₂ (CPM + 1): *Atoh1* = *Slc17a6*⁺/*Meis2*⁺/*Lmx1b*⁻, *Lmx1* = *Slc17a6*⁺/*Meis2*⁻/*Lmx1b*⁺, inhibitory = GABAergic neurons, and other = remaining neurons. Target labels indicate c-Fos upregulation in the Control < Isoflurane UNRAVEL analysis: bold, FDR $q < 0.005$; regular black, $0.005 \leq q < 0.05$; gray, not detected at $q < 0.05$. Nearly all LPB target regions shown had increased c-Fos, suggesting that isoflurane-activated glutamatergic LPB neurons may contribute to widespread downstream activity. **(F)** Imputed marker expression across ABCA neurons within the LPB portion of cluster 3. *Slc17a6* encodes VGLUT2, *Slc32a1* encodes VGAT, *Chat* encodes ChAT, and *Slc6a5* encodes GlyT2. The transcription factors *Lmx1a* and *Lmx1b* are expressed in the *Lmx1* clade, whereas *Evx2* and *Meis2* are expressed in the *Atoh1* clade. Examples of neuropeptides preferentially associated with *Atoh1*- or *Lmx1*-related LPB populations are shown. Region abbreviations in **(E)**: AI, agranular insula; BNST, bed nucleus of the stria terminalis; CeA, central amygdala; CM, central medial thalamus; DMH, dorsomedial hypothalamus; DR, dorsal raphe; IMD, intermediodorsal thalamus; LHA, lateral hypothalamic area; LS, lateral septum; MEPO, median preoptic nucleus; NTS, nucleus of the solitary tract; PAG, periaqueductal gray; PSTN, paraventricular nucleus; PVH, paraventricular hypothalamus; RE, nucleus of reuniens; RN, reticular nucleus; SI, substantia innominata; VMH, ventromedial hypothalamus; VPMpc, ventral posteromedial thalamus, parvocellular; VTA, ventral tegmental area.